

Positional Mechanisms in Rotary Attention: two factors, one scan

Notes derived from the implementation

Abstract

Rotary attention writes every logit as a sum of products $\mu \cos \varphi$. **A positional mechanism is a choice about one of those two factors, and nothing else.** MapFormer’s path integration, Selective RoPE and ordinary RoPE all set the *phase* φ ; PoPE sets the *magnitude* μ ; MapPoPE is their conjunction and not a third mechanism.

The phase mechanisms are one machine. Each maintains a state by cumulative summation and reads it out linearly into the phase, $s_t = s_{t-1} + \varphi(x_{t-K+1:t})$, $\theta_t = As_t$, and they differ only in the *state rank* m , the nonlinearity of the increment, and its temporal support K . RoPE, MapFormer and Selective RoPE are then *nested*, each inclusion removing a constraint rather than adding a mechanism, so every comparison between them asks whether a constraint is worth keeping and none of them is about expressivity. For MapFormer the reduction is an identity: all 64 of its phase channels are linear readouts of a single *two*-dimensional accumulator (in this implementation, whose bottleneck is shared across heads; in the paper’s per-head design each head reads its own two-dimensional accumulator).

Rank is the axis that binds, and it binds through optimisation, not geometry. Confining the phase to an r -dimensional subtorus forces the N^D positions of a D -torus apart by no more than $\sim N^{-\max(D/r, 1)}$; the bound is correct as algebra but does not account for trained models ($r = 2$ is unpaired at $D = 5$). The measured cost of $r = 2$ in two dimensions is a *search* failure: trained and tested at $T = 1024$, $r = 2$ usually fails to find a solution that a rank-2 projection of a solved $r = 4$ model shows exists and holds under training. §2.5 states both, with the correction that the first version of the geometric argument — an appeal to a kernel — was wrong and was falsified by its own gate.

The asymmetry between the two factors matters. MapFormer adds a term to the phase and leaves the rest of attention untouched. PoPE replaces the magnitude *and* eliminates the content-dependent intrinsic phase that ordinary rotary attention carries, so it is a *restriction* on the object MapFormer rotates rather than a parallel choice — a plausible source of the sub-additivity seen on text (§4).

Formulas are transcribed from the implementation. Where it departs from a published equation or from canonical RoPE, that is stated at the point of departure. Empirical claims carry their seed counts and minimum detectable effects; several are recorded here only to be marked unsupported.

Corrected 2026-09-25. This record predates three lines of work, and where its body disagrees with them they win. (i) The EM/WM audit (AUDIT_2026-09-10.md): MapWM rotates content Q, K and is *not* additive, Thm 3 is withdrawn, and the sign of the separate- q_0/k_0 effect depends on the task. (ii) The PoPE line: the accumulator-plus-kernel account of the MapPoPE collapse and its PoPE-decoupling corollary are withdrawn (T2.RESULTS.md), “PoPE’s Table 5 does not replicate” is withdrawn (ABLATE.RESULTS.md), and “PoPE is inert with an index phase” was a 16-epoch artefact (PAPER2X2.RESULTS.md). (iii) Rank at matched length (RANK_MI.RESULTS.md, RANK_PROJ.RESULTS.md): the rank deficit is search, not a skewed basis; and the torus numbers $+0.461 / +0.003$ quoted below are the 16-epoch recipe, converged $+0.243$ with the encoding detectable at every length.

1 The shared frame

Let $x_1, \dots, x_T \in \mathbb{R}^d$ be token embeddings, and for a single head write queries and keys $q_t = W_Q x_t$, $k_s = W_K x_s \in \mathbb{R}^{d_{\text{head}}}$. Group the d_{head} coordinates into n_b two-dimensional blocks and identify each block with a complex number,

$$\tilde{q}_t^{(c)} = q_{t,2c} + i q_{t,2c+1}, \quad \tilde{k}_s^{(c)} = k_{s,2c} + i k_{s,2c+1}, \quad c = 0, \dots, n_b - 1. \quad (1)$$

A rotary scheme attaches a per-position angle $\theta_t^{(c)}$ and rotates:

$$\begin{pmatrix} q'_{t,2c} \\ q'_{t,2c+1} \end{pmatrix} = \underbrace{\begin{pmatrix} \cos \theta_t^{(c)} & -\sin \theta_t^{(c)} \\ \sin \theta_t^{(c)} & \cos \theta_t^{(c)} \end{pmatrix}}_{R(\theta_t^{(c)}) \in SO(2)} \begin{pmatrix} q_{t,2c} \\ q_{t,2c+1} \end{pmatrix}, \quad (2)$$

and likewise for k_s . The unnormalised attention logit is then

$$a_{ts} = \frac{1}{\sqrt{d_{\text{head}}}} \langle q'_t, k'_s \rangle = \frac{1}{\sqrt{d_{\text{head}}}} \sum_{c=0}^{n_b-1} \langle R(\theta_t^{(c)}) q_t^{(c)}, R(\theta_s^{(c)}) k_s^{(c)} \rangle. \quad (3)$$

Because R is orthogonal, $R(\alpha)^\top R(\beta) = R(\beta - \alpha)$, so

$$a_{ts} = \frac{1}{\sqrt{d_{\text{head}}}} \sum_c \langle q_t^{(c)}, R(\theta_s^{(c)} - \theta_t^{(c)}) k_s^{(c)} \rangle = \frac{1}{\sqrt{d_{\text{head}}}} \sum_c \text{Re} \left[\overline{\tilde{q}_t^{(c)}} \tilde{k}_s^{(c)} e^{i(\theta_s^{(c)} - \theta_t^{(c)})} \right]. \quad (4)$$

Only the *difference* of angles survives: this is the relative-position property, and it holds for *any* choice of θ .

Finally, write each complex factor in polar form, $\tilde{q}_t^{(c)} = \mu_{t,c}^q e^{i\psi_{t,c}^q}$ and similarly for k . Then (4) becomes

$$a_{ts} = \frac{1}{\sqrt{d_{\text{head}}}} \sum_c \underbrace{\mu_{t,c}^q \mu_{s,c}^k}_{\text{magnitude}} \cos \left(\underbrace{\theta_s^{(c)} - \theta_t^{(c)} + \psi_{s,c}^k - \psi_{t,c}^q}_{\text{phase}} \right). \quad (5)$$

Equation (5) is the frame for everything that follows. **Each mechanism is a decision about where the magnitude comes from and where the phase comes from.**

2 The phase axis: a family of scans

Every mechanism that sets θ in (5) does so the same way. Call a **phase generator** the triple (m, φ, K) producing

$$s_t = s_{t-1} + \varphi(x_{t-K+1}, \dots, x_t) \in \mathbb{R}^m, \quad \theta_t = A s_t \in \mathbb{R}^{H n_b}. \quad (6)$$

m is the **state rank**, φ the increment map, K its temporal support and A a fixed linear readout. Cumulative summation is linear, so it commutes with any linear map: the accumulation may always be pushed down to the smallest space the increment factors through, and m is that dimension. Three named mechanisms are instances, and §2.1–2.3 give them in turn.

generator	m	φ	K	A
RoPE	1	$\equiv 1$	1	ω (frozen)
MapFormer	r	$W_{\text{in}}x$	1	$\text{diag}(\omega) W_{\text{out}}$
Selective RoPE	HN_b	$\sigma(W_g x) \odot (\kappa * W_\omega x)$	$4^\dagger$	τI

[†] our choice, not theirs: the paper states no convolution width, and its own derivation implies $K = 2$. See §2.6.

Because the state is accumulated and the readout is linear, all three are $O(\log T)$ parallel scans. That property is shared and is not what distinguishes them.

2.1 RoPE: the fixed clock

Standard RoPE fixes the angle to be linear in the token index,

$$\theta_t^{(c)} = t \cdot \omega_c, \quad \omega_c = \text{base}^{-2c/d_{\text{head}}} = \text{base}^{-c/n_b}, \quad (7)$$

with ω a non-learnable buffer. The magnitude and the intrinsic phase ψ are whatever W_Q, W_K happen to produce; they are never constrained, and in particular $\tilde{q}$ may lie anywhere in $\mathbb{C}$. Substituting into (4) gives the familiar dependence on $s - t$ alone.

2.2 MapFormer: path integration replaces the clock

MapFormer keeps (2) and changes only what generates θ .

Step 1 — token to Lie-algebra increment. A low-rank map (rank $r = 2$; paper App. A.7, $W_\Delta = W_\Delta^{\text{out}} W_\Delta^{\text{in}}$) reads *every* token, action or observation:

$$\Delta_t = W_{\text{out}} W_{\text{in}} x_t \in \mathbb{R}^{H \times n_b}, \quad W_{\text{in}} \in \mathbb{R}^{r \times d}, \quad W_{\text{out}} \in \mathbb{R}^{(H n_b) \times r}. \quad (8)$$

Nothing tells the model which tokens are actions; it must *learn* $\Delta \approx 0$ on observations. The paper states only $r \ll d$, “for instance $r = 2$ ” (App. A.7).

Superseded. An earlier note here recorded the bottleneck as load-bearing with $r = 1$ dropping the 2D task to 0.66, and marked it unsupported. *Corrected 2026-09-05:* “no $r = 1$ run exists anywhere” was wrong about the paper. The locally held copy is **v3** (6 Feb 2026), which has no rank-split arms; **v4** splits MapWM into r1/r2 and reports an r1 row, so the 0.66 figure is most likely the paper’s own 2D IID number rather than folklore. Treat it as the paper’s evidence for the structural rank argument, not as unsupported — but the rank axis is no longer untested: §2.5 reports an 8-seed sweep over $r \in \{2, 4, 8, 16, 32\}$ in which $r = 2$ is the *worst* setting. Read $r = 2$ below as the paper’s prescription, not as a verified optimum.

Step 2 — integrate.

$$\boxed{\theta_t^{(h,c)} = \omega^{(h,c)} \sum_{u \leq t} \Delta_u^{(h,c)} = \omega^{(h,c)} \text{cumsum}(\Delta)_t^{(h,c)}}. \quad (9)$$

Because $SO(2)$ is abelian, the prefix product of rotations collapses to a *cumulative sum of angles*, which is what makes this an $O(\log T)$ parallel scan rather than a sequential recurrence:

$$\prod_{u \leq t} R(\omega \Delta_u) = R\left(\omega \sum_{u \leq t} \Delta_u\right). \quad (10)$$

Commutativity is the entire computational argument, and it is also the limitation: a turn/forward action space, where displacement depends on accumulated heading, is *not* representable by (9).

Step 3 — frequency schedule. The ω are learnable, initialised geometrically over n_b blocks:

$$\omega_i = \omega_{\max} \left(\frac{1}{\Delta_{\max}} \right)^{\frac{i}{n_b-1}}, \quad i = 0, \dots, n_b - 1, \quad \omega_{\max} = 2\pi, \quad \Delta_{\max} = \text{grid size}. \quad (11)$$

Correction to the published equation. Eq. (11) differs from the paper (v1/v3 eq. 17, v4 eq. 18) in *two* places, both deliberate. First the **sign**: the paper prints a negative exponent, giving an *increasing* schedule that contradicts its own stated boundary condition $\omega_{\min} = \omega_{\max}/\Delta_{\max} \leq \omega_i \leq \omega_{\max}$. Second the **denominator**: the paper writes n_b with $1 \leq i \leq n_b$, so with the sign defect repaired the endpoint $\omega_{\max}$ is never attained (at $i = 1$) while $\omega_{\min}$ is; eq. (11) uses $n_b - 1$ with $0 \leq i \leq n_b - 1$, which hits both endpoints exactly. The corrected form is what satisfies the paper’s own inequality and what matches RoPE’s $\theta_i = \text{base}^{-2i/d}$. Both defects survive into v4.

RoPE is a special case — of the functional form, not the encoding. If the model learns $\Delta_t \equiv 1$ for all t , (9) becomes $\theta_t = (t+1)\omega$ (the cumsum is inclusive; only differences enter (4), so the offset is immaterial). That is RoPE’s *form*, with the token index as the clock. It is *not* RoPE identically, for three reasons: ω here is a trained parameter rather than a frozen buffer; its schedule spans $[2\pi/64, 2\pi] = [0.098, 6.28]$ against RoPE’s $[1.3 \times 10^{-4}, 1]$, a ratio running from $6.3\times$ at $c = 0$ to $736\times$ at $c = n_b - 1$; and Δ_t is a rank-2 function of the embedding, so $\Delta \equiv 1$ is a hypothesis about what was learned, not a setting. The accurate statement is that path integration is **RoPE with a learned, content-dependent clock rate — which may be zero or negative — over a learned frequency spectrum.**

In terms of (5): MapFormer sets the **phase** and leaves the magnitude untouched.

2.3 Selective RoPE: the unconstrained generator

Selective RoPE (Movahedi et al., arXiv:2511.17388, ICLR 2026) was posted on 21 November 2025, three days before MapFormer, and neither paper cites the other. (Dates from the arXiv identifiers; the mutual non-citation is verified in the versions held locally, SRoPE v3 and MapFormer v4, so “in any version” is more than two PDFs can establish.) Its angle is, in the present notation,

$$\theta_t^{(h,c)} = \tau \sum_{u \leq t} \left[\underbrace{\sigma(W_g x_u)}_{\text{gate}} \odot \underbrace{(\kappa * W_\omega x)_u}_{\text{smoothed increment}} \right]^{(h,c)}, \quad (12)$$

with κ a causal depthwise kernel and τ a learned temperature. (We take the width to be 4; the paper states none, and §2.6 shows why that matters more than it looks.) It is (6) at $m = Hn_b$: **64 channels integrating 64 independent signals**, against MapFormer’s 64 readouts of one two-dimensional accumulator. Their *baseline* has no rank bottleneck, so no cross-channel constraint, and $K > 1$.

Their stability fixes converge on ours. The paper reports that Selective RoPE trains unstably at higher learning rates and that this is remedied “by either placing a weight norm on the ω projection or using a low-rank MLP with rank 8 or 16”. A documented variant of Selective RoPE is therefore *low-rank*, at a rank in the same range this project measures as sufficient. Calling it “the unconstrained generator” is accurate for the arm implemented here and for their headline design, and overstates the distance between the two papers.

Scope of every Selective-RoPE number below. The arms here swap the *generator* and keep MapFormer’s *placement* — the angle is computed once from the token embeddings before the blocks, not per head and per layer from that layer’s queries. At one layer the distinction is

nearly vacuous, since $q = W_Q \text{LN}(x)$ is itself a learned linear map of the token. At depth it is not, and §10.4 records it as the one axis on which the two designs remain uncomparred. *Correction:* an earlier version of this box also called the token-versus-query source a deviation. It is not — their own ablation moves to the input: “we find that this can be remedied by using the ℓ_2 -normalised input x_t , instead of the queries q_t , to parametrise the rotation angles ω ”. Only the per-layer half of the placement gap is real. **A negative result here does not refute Selective RoPE**; it is evidence about the generator’s components in this setting, on tasks their paper does not run.

2.4 The inclusion

The three generators are nested, and each inclusion *removes a constraint* rather than adding a mechanism:

$$\underbrace{\text{RoPE}}_{m=1, \varphi \equiv \text{const}} \subset \underbrace{\text{MapFormer}}_{m=r, \varphi \text{ linear}, K=1} \subset \underbrace{\text{Selective RoPE}}_{m=Hn_b, \varphi \text{ gated}, K>1}.$$

The right-hand inclusion is **exact**, and it is worth constructing rather than asserting. Given MapFormer’s $(W_{\text{in}}, W_{\text{out}}, \omega)$, set Selective RoPE’s parameters as follows.

- 1 $\kappa :=$ the delta at the *last* tap, $(0, 0, 0, 1)$. The convolution is left-padded by $K - 1$, so $\text{out}_t = \sum_j w_j u_{t+j-3}$ and this weight gives $\text{out}_t = u_t$: K **collapses** $4 \rightarrow 1$.
- 2 $W_g := 0$. The gate is then the *constant* $\sigma(b_g) \in (0, 1)^{Hn_b}$ for *any* bias, and a constant is absorbed at step 3.
- 3 $W_\omega := \text{diag}(\text{vec } \omega) W_{\text{out}} W_{\text{in}} / (\tau \sigma(b_g))$, row-wise. Legal because the full-rank parameterisation does not *constrain* rank, and weight normalisation represents any matrix (rows $g_i v_i / \|v_i\|$, with $g_i = 0$ for a zero row).
- 4 $b_\omega := 0$, or MapFormer acquires an index-RoPE component.

Verified against both implementations (`verify_inclusion.py`): $\max |\Delta \cos| = 1.5 \times 10^{-5}$ over a length-64 stream.

The reduction, line by line. Substituting the four settings into (12) and using that cumsum is linear:

$$\begin{aligned} \theta_t &= \tau \sum_{u \leq t} \left[\sigma(W_g x_u) \odot (\kappa * W_\omega x)_u \right] && \text{Selective RoPE, (12)} \\ &= \tau \sum_{u \leq t} \left[\sigma(b_g) \odot (\kappa * W_\omega x)_u \right] && \text{step 2: } W_g = 0 \\ &= \tau \sigma(b_g) \odot \sum_{u \leq t} W_\omega x_u && \text{step 1: } \kappa = \delta; \text{ constants leave the sum} \\ &= \tau \sigma(b_g) \odot \frac{\text{diag}(\text{vec } \omega) W_{\text{out}} W_{\text{in}}}{\tau \sigma(b_g)} \sum_{u \leq t} x_u && \text{step 3} \\ &= \omega \odot W_{\text{out}} W_{\text{in}} \sum_{u \leq t} x_u = \omega \odot \text{cumsum}(\Delta)_t && \text{which is (9).} \end{aligned} \tag{13}$$

Each line is an identity, not an approximation, which is what makes the inclusion exact rather than asymptotic.

Correction. An earlier version of this paragraph called the gate step a *saturating limit*. It is not a limit: $W_g = 0$ makes the gate exactly constant for any bias, and the constant is absorbed. The inclusion is exact, not asymptotic.

The inclusion is on the function, not the state. Selective RoPE realises MapFormer’s θ with a 64-dimensional state confined to an r -dimensional subspace: the constructed W_ω has rank 2, measured. This forces the definition in §2.5 to be taken on the **composite** token-to-phase map $A \cdot B$ where $\varphi(x) = Bx$, not on A alone — otherwise this very construction would read $\rho = Hn_b = 64$ while computing a rank-2 function.

The chain is not uniformly exact. RoPE sits inside Selective RoPE *unconditionally*: $W_\omega := 0$ with bias b gives $\theta_t = \tau(t+1)b$, an exact index clock (constant to 7×10^{-9}). It sits inside MapFormer at a **price**, because Δ has **no bias term**: $\Delta_t \equiv \text{const}$ requires every embedding row to share its projection onto W_{in} . *Corrected on audit*: that is a setting one can choose, not a hypothesis about training. Take $W_{\text{in}} = [I_r \ 0]$ and pin the first r embedding coordinates to a common value; Δ is then constant across tokens to exactly 0 and $\theta_t = (t+1)\omega \odot \Delta$ reproduces to 3.5×10^{-6} . The cost is r of the d embedding dimensions — at $V = 21$, `token_emb` remains full rank, so on this task it costs nothing. Inside Selective RoPE the same special case is a setting of the generator alone. The three further qualifications are in §3.

Consequently **MapFormer’s generator is a constrained submodel of Selective RoPE’s**, and every comparison between the two is a question about whether a constraint is worth keeping. It is never a question about expressivity, and no experiment here can be read as one.

2.5 Coherence rank, and the two thresholds

Define the **coherence rank** as the rank of the *composite* token-to-phase map, $\rho = \text{rank}(A \cdot B)$ where $\varphi(x) = Bx$: the dimension of the *affine* span of $\{\theta_t\}$, i.e. $\text{rank}[A \cdot B \mid A\varphi(0)]$. The affine part is not decoration — RoPE’s $\varphi \equiv 1$ is constant, so $B = 0$ and the linear rank alone would read 0 for the one case the next sentence asserts. It is 1 for RoPE, r for MapFormer, and generically Hn_b for Selective RoPE. Taking the rank on A alone would be wrong — §2.4 constructs a Selective RoPE with $A = \tau I$ that computes a rank-2 function.

For MapFormer this is an identity rather than a reading. Because $\Delta_t = W_{\text{out}}W_{\text{in}}x_t$ and cumsum is linear,

$$\theta_t = \omega \odot \text{cumsum}(W_{\text{out}}W_{\text{in}}x)_t = \omega \odot W_{\text{out}} \underbrace{\sum_{u \leq t} W_{\text{in}}x_u}_{s_t \in \mathbb{R}^r}, \quad (14)$$

checked against the implementation at 1.5×10^{-5} in float32 (`verify_theory.py`). **All $Hn_b = 64$ phase channels are linear readouts of a single two-dimensional accumulator.**

Rank binds twice, for two unrelated reasons, and separating them is the point of this section.

(i) Geometry: a packing bound, with a correction

The first version of this argument was wrong, and its own gate falsified it. It claimed that on a D -dimensional lattice the induced map $\mathbb{Z}^D \rightarrow \mathbb{R}^r$ has a kernel of dimension $D - r$, so that $r < D$ makes distinct positions *identical* in phase. That is false. The kernel of a generic linear map is a $(D - r)$ -dimensional subspace of $\mathbb{R}^D$ containing no non-zero lattice point almost surely, so a generic W is injective on any finite set of positions and exact collisions never occur. The validator written to measure the collision rate returned 0.0% at every (D, r) , which is what a correct gate on a wrong claim looks like. What actually binds is not identity but *separation*, below.

The phase enters attention only through $\cos(\theta_s - \theta_t + \psi)$, which is 2π -periodic in each block, so the code lives on the torus $\mathbb{T}^{Hn_b}$ — and by (6) it is confined to an ρ -dimensional subtorus of it. Packing the N^D positions of a D -torus into an ρ -dimensional one forces them close together: with the cloud normalised to unit scale, the minimum separation between distinct positions obeys

$$\min_{p \neq p'} \|s(p) - s(p')\| \lesssim N^{-\max(D/\rho, 1)}. \quad (15)$$

Two regimes follow, and the exponent is the whole content of the bound.

- $\rho \geq D$: the accumulator can embed the position lattice faithfully, and the separation is set by the lattice’s own minimal distance against a spread of order N . The exponent **saturates at 1**; more rank buys nothing geometric.
- $\rho < D$: the image is a projected lattice, dense in the subtorus, and the exponent is $D/\rho > 1$ — strictly worse, and worse the smaller ρ is.

Measured by projecting the full position set through random rank- ρ maps and fitting $\log(\text{separation})$ against $\log N$:

D	ρ	D/ρ	predicted exponent	fitted
2	2	1.00	−1.00	−1.02
2	4	0.50	−1.00 (<i>saturated</i>)	−1.04
3	2	1.50	−1.50	−1.51
3	3	1.00	−1.00	−1.44
4	2	2.00	−2.00	−2.17
4	4	1.00	−1.00	−1.23

Two corrections, both found on audit. (a) *These exponents are ladder-dominated.* Each is fitted over four map sizes, and the fitted value moves with which four: the repo’s own gate (ND_GATES.md), run on its capped ladder, reports −1.04 where this table reports −1.51 for the same $D=3, \rho=2$ cell (and −1.40 against −2.17 at $D=4, \rho=2$). Re-measured on a wide ladder with 60 projections per point, five of the six cells reproduce (e.g. −2.18 against the printed −2.17), so the table is not wrong — but at $D=5$ the measured exponents are −2.99 against a predicted −2.50 and −1.18 against −1.00, so the bound is loose there and “the exponent is verified” is not earned. The bound is an upper bound and no cell violates it. (b) *The note elsewhere reads the bound through its CONSTANT, not its exponent,* and the two readings disagree. Under the exponent reading geometry predicts nothing anywhere in the §10.3 design, since every arm has $\rho \geq 2$ and the exponents are flat. The constant reading was then used to withdraw the separability claim outright — **and that withdrawal was itself wrong.**

(b') *The withdrawal is retracted, on a second audit.* It rested on the separations 0.033/0.065/0.193 printed for $r = 2/3/5$ at $D=3$, which come from `validate.nd.py::g5_separation` with `trials=5`. A minimum distance is an extreme-value statistic and five random projections do not estimate one: at $n = 200$ the same quantity is 0.028 ± 0.015 , 0.108 ± 0.053 , 0.181 ± 0.059 — relative spreads near 50%, and the pivotal $r = 3$ cell off by 65%. The **ordering inverts**: the $r=2 \rightarrow r=D$ ratio is $\approx 3.9\times$ and the $r=D \rightarrow r=D+2$ ratio $\approx 1.7\times$, so the first jump is more than twice the second — which is what the separability design needs, not what defeats it. The design table’s ratio row is affected the same way. So the sentence in §10.3 about what geometry predicts stands; what falls is the five-sample estimator, and every number in this project that came out of it.

TESTED, AND IT DOES NOT BIND (§10.3). The bound is correct as algebra; its exponent is *not* verified here (see the audit note under the table — the fitted ladder has no

surviving script, and for the one cell it shares with `ND_GATES.md` the two disagree, -1.51 against -1.04). At $n = 8$ across $D \in \{2, 3, 5\}$ two of the three pre-registered falsifiers fired: $r = 2$ is not impaired at $D = 5$ — it posts its *highest* score there (0.896 at $T=512$, against 0.848 at $D=2$ and 0.707 at $D=3$) — and the $r = 2$ deficit does not grow with D . Read what follows as a true statement about the geometry of the code that is not the reason the code fails.

What this does *not* explain. At $D = 2$ both $\rho = 2$ and $\rho = 4$ sit at the floor exponent, so the bound predicts *no* advantage for $r = 4$ on a two-dimensional grid. One is measured, at $+0.085$ and $8/8$ seeds. The geometric argument is therefore silent about the effect this project actually observed, which is the reason for (ii) — and a useful discipline, since a theory that explained the 2D gap as well would have been explaining it twice.

(ii) Optimisation: $\rho = D$ is expressible but not reachable

In 2D, $r = 2$ is exactly sufficient by (i) and is what the paper prescribes (r is the dimensionality of the action space, App. A.7) — *per head*: the paper’s $W_{\text{in}} \in \mathbb{R}^{d \times n_h \times r}$, while this implementation shares one r -dimensional latent across heads, so at two heads our $r = 2$ has half the paper’s latent dimensions. Trained at $T = 128$, it nonetheless costs 0.085 accuracy at $T = 1024$ ($8\times$ the training length) against $r = 4$ ($8/8$ seeds, $t = 3.57$), and on average the code it learns is skewed: opposite actions fail to cancel by half the action scale (0.495) and the two axes are close to parallel ($|\cos| = 0.783$), against 0.092 and 0.175 at $r = 4$. Yet the $r = 4$ latents still place 100.0% of their spectral energy in a 2-plane, and $r = 8, 16, 32$ are flat within noise of $r = 4$. Within $r = 2$ the skew does not predict accuracy ($r = -0.35$ and $+0.30$ for the two metrics), so it describes the arm and is not the cause.

Tested at matched length (2026-09-24), and the heading above holds. Trained and tested at $T = 1024$ for 900 epochs, every arm built from the same initial weights, 8 seeds: our shared $r = 2$ solves $0/8$ (accuracy 0.894), the paper’s per-head $r = 2$ $2/8$ (0.885), a shared $r = 4$ $8/8$ (0.998); Fisher $p = 0.0002$ and 0.007 against $r = 4$. The per-head arm and $r = 4$ have the same four latent dimensions, so what decides it is the rank of each head’s content-to-angle map. A rank-2 projection of each solved $r = 4$ model scores 0.9955 frozen, and trained onward it stays solved ($7/8$, as does the $r = 4$ control). The solution is expressible and stable at rank 2 and training from scratch does not find it. Budget-scoped (within 900 epochs), one task and width, and $r = 4$ ’s smaller initial W_{out} (bound 0.5 against 0.707) is not separated from its rank (`RANK_MI_RESULTS.md`, `RANK_PROJ_RESULTS.md`).

The extra rank carries no signal; it is slack that lets gradient descent reach an exact group homomorphism it otherwise misses. The representation is two-dimensional at convergence either way. The paper’s structural argument for $r = 2$ is right about what is expressible and wrong about what is reachable, and the distinction is not rhetorical — it is the difference between a bound that saturates at $r = D$ and a measurement that does not.

Why coherence is cheap to keep

Nothing above says a *large* ρ is harmful in itself, and the direct comparison bears that out: the unconstrained generator’s $\rho = 64$ buys no more on the torus than $r = 4$ does (§10), at $20\times$ the parameters. The constraint is worth keeping because it is nearly free and because it is what makes the phase a multi-scale code for one quantity rather than 64 unrelated clocks — not because incoherence has been shown to cost. On parity, where the latent is a single bit and $\rho^* = 1$, the unconstrained generator does measurably worse; that is the only place in these experiments where incoherence has a price.

2.6 What $K > 1$ costs: the group structure

With $K = 1$ and φ linear, (6) is exactly the action of the free commutative monoid on tokens: the phase depends on the *multiset* of tokens traversed and on nothing else, so $\theta(a \text{ then } b) = \theta(b \text{ then } a) = \theta(a) + \theta(b)$, and $\prod_{u \leq t} R(\omega \Delta_u) = R(\omega \sum_{u \leq t} \Delta_u)$. Commutativity is MapFormer’s entire computational argument, and it is also its stated limitation: a turn/forward action space, where displacement depends on accumulated heading, is not representable — measured on the torus with rotation actions, as -0.388 of a -0.438 swing attributable to rotation alone.

A convolution with $K > 1$ makes the increment of a token depend on its neighbours, so the phase ceases to be a function of the multiset and the map is no longer a homomorphism. The word “commutativity” does not appear once in Selective RoPE. The measured cost of giving it up is -0.020 on parity and -0.064 on the torus, for 193 parameters — the only knob in §10 that is free and negative on both tasks.¹ Its *torus* sign was pre-registered; its parity sign was not — the pre-registration was, in SELECTIVE_ROPE.md’s own words, “half right ... it hurts on parity too, which the mechanism story did not predict”.

The K axis is sharper than “blurring”, and their own derivation says so. Selective RoPE motivates its convolution as a *shift*: “the shift over the queries q can be expressed by a 1d short-convolution”, from an angle $\phi_j = \langle \omega_j, q_t - q_{t-1} \rangle$. That is a first difference, $K = 2$ with weights $(-1, +1)$ — and a first difference inside a cumulative sum *telescopes*:

$$\sum_{u \leq t} (z_u - z_{u-1}) = z_t - z_0. \quad (16)$$

With the exact difference kernel the angle is a function of the *current* token alone and no accumulation is left. So K does not interpolate between sharp and blurred increments; it interpolates between **accumulation** ($K = 1$, identity kernel) and **no accumulation** (the differencer), and the kernel’s DC gain says where a trained model sits.

Measured, and the answer is neither (CONV_KERNEL_PROBE.md). Projecting the learned unit-normalised kernels onto the identity and onto the differencer gives 0.423–0.470 and 0.345–0.463 across both tasks, against 0.424 for a kernel drawn uniformly at random. The convolution does not learn to get out of the way, and does not learn the difference its own derivation motivates — it stays an arbitrary smear of the increment applied before accumulation, which is a sufficient account of the cost. **The width $K = 4$ used here is an assumption**, following the Mamba/GLA convention; the paper states no width in its pseudocode, its two ablation tables or its hyperparameter appendices, and $K = 2$ with a fixed difference kernel — the setting its derivation specifies — has not been run.

3 The magnitude axis: PoPE

PoPE (Polar Coordinate Positional Embeddings) makes (5) explicit rather than emergent.

Notation change, for this section and the next. §1 defined n_b as the number of $2D$ blocks, $n_b = d_{\text{head}}/2$. PoPE indexes one frequency per *element*, so throughout this section and the next

¹Read every per-knob number in this document against the confound recorded in §10. The **SelectiveAngle** arms are not single-knob: none carries the ω buffer, so each also swaps the readout $\text{diag}(\omega)W_{\text{out}} \rightarrow \tau W_2$. For the conv and gate rows this preserves the function class exactly and is an *initialisation* difference (geometric ω against a default draw) plus one scalar; for the two full-rank rows the class genuinely differs, since they also add **weight_norm** and a bias. So -0.020 and -0.064 are attributable up to initialisation, the $\rho = 64$ comparisons are not, and the arm that would settle both — geometric ω retained, one knob added — has not been run.

$n_b = d_{\text{head}}$. Read $\vartheta_c = \text{base}^{-c/n_b}$ below as $\text{base}^{-c/d_{\text{head}}}$; under §1’s definition it would be wrong by a factor of two in the exponent.

Per element c :

$$\mu_{t,c}^q = \text{softplus}(Q_{t,c}), \quad \mu_{s,c}^k = \text{softplus}(K_{s,c}), \quad (\text{content, strictly positive}) \quad (17)$$

$$\varphi_{t,c}^q = t \vartheta_c, \quad \varphi_{s,c}^k = s \vartheta_c + \delta_c, \quad (\text{position, plus a bias}) \quad (18)$$

with $\vartheta_c = \text{base}^{-c/n_b}$ a **fixed buffer** and δ_c a learnable per-(head, frequency) constant.

δ_c is **very nearly inert in practice**. It is clamped to $[-2\pi, 0]$ in the *forward pass*, not projected onto that range, so its gradient is identically zero outside the interval — and it is initialised at exactly 0, i.e. *on* the upper boundary. Any step that would push it positive kills its gradient permanently. Measured on trained checkpoints (4 heads $\times$ 64 frequencies $\times$ 4 layers, $n = 36$), **83–98% (mean 93%) of δ_c entries end at ≥ 0** , clamped to zero with dead gradient; per-layer means run -0.0003 to -0.018 rad. For roughly 95% of frequencies, (19) is effectively $\cos((s-t)\vartheta_c)$. Treat the bias as a nominal degree of freedom, not a working one.

The logit is

$$a_{ts} = \frac{1}{\sqrt{d_{\text{head}}}} \sum_c \text{softplus}(Q_{t,c}) \text{softplus}(K_{s,c}) \cos((s-t)\vartheta_c + \delta_c). \quad (19)$$

Implemented via the angle-difference expansion, so no complex arithmetic is needed:

$$\cos_{s,c}^K = \cos \varphi_{s,c} \cos \delta_c - \sin \varphi_{s,c} \sin \delta_c, \quad \sin_{s,c}^K = \sin \varphi_{s,c} \cos \delta_c + \cos \varphi_{s,c} \sin \delta_c, \quad (20)$$

$$a_{ts} = \frac{1}{\sqrt{d_{\text{head}}}} \left[(\mu^q \cos \varphi)_t \cdot (\mu^k \cos^K)_s^\top + (\mu^q \sin \varphi)_t \cdot (\mu^k \sin^K)_s^\top \right]. \quad (21)$$

What PoPE does *not* do. Its phase is a function of position only. The magnitude is content-dependent; the *angle* is not, apart from the constant bias δ_c . This is worth stating because the name invites the opposite reading: PoPE is **not** in the CoPE / Selective-RoPE family, which learn content-dependent *angles*. It modifies the other factor.

In terms of (5): PoPE sets the **magnitude** and leaves the phase as a fixed clock.

3.1 The magnitude axis is also a scan: forget gates

Selective RoPE’s own framing arrives at the two-factor decomposition from the opposite direction, and it is worth recording because it names a cell this note had left empty. Their stated design principle is *rotation and decay*: recall needs “(i) rotation, to encode relative position while preserving norms, and (ii) decay, to selectively discard past key–value associations”, summarised as **“Real parts forget; imaginary parts encode position.”** A complex diagonal state transition has a modulus and a phase; the modulus is a forget gate, the phase is a rotation.

That is (5)’s two factors, derived from diagonal SSMS rather than from the polar form of a rotary logit. But their magnitude is *accumulated* — a product $\prod_u \gamma_u$ over the transition — where PoPE’s is *instantaneous*, a function of the current token only. Crossing that against the phase axis completes the design space:

	phase	magnitude
accumulated	MapFormer, Selective RoPE	forget gate / decay (GLA, Mamba, FoX)
instantaneous	ψ (free in RoPE; deleted by PoPE)	PoPE’s softplus(Q)

MapFormer occupies one cell and has nothing in the accumulated-magnitude one. It uses softmax attention, which carries no decay, and adds no forget gate. On their reading that is a real omission rather than a stylistic difference — their classification of existing models runs the same way, with DeltaNet already possessing a rotational component and gaining from an added forget gate, and the softmax Transformer possessing rotation implicitly along random axes so that “only a forget gate is needed to fully satisfy the rotation + decay recipe”, which is the Forgetting Transformer.

Which of their selectivities is MapFormer’s. *Selective rotation*, exactly: same factor, same accumulation, differing only in the (m, φ, K) settings of §2. *Selective decay* has no MapFormer counterpart at all. Whether it should is untested here — a forget gate on a navigation task with a bounded map is not obviously wanted, since discarding old key–value associations is the opposite of what a cognitive map is for.

Does anything on their side resemble MapEM? MapEM computes two attention matrices and multiplies them elementwise, $A = \text{softmax}(A_X \odot A_P)$. Nothing in Selective RoPE does that, and the closest structural analogue is the forget gate rather than anything in the rotation. Writing their eq. (13) with a decayed transition, the weight linking query t to key τ factors as

$$\underbrace{\left(\prod_{u=\tau+1}^t \gamma_u \right)}_{\text{positional}} \times \underbrace{k_\tau^\top \check{R}_{\tau+1:t} q_t}_{\text{content} \times \text{position}}, \quad (22)$$

which has the *shape* of $A_X \odot A_P$ — a positional scalar gating a content term, an AND rather than an OR. Three things make it a weak analogue. The positional factor is a *monotone* decay, so it can only downweight with distance and can never peak at a particular displacement, which is the thing a cognitive map needs. It is itself content-dependent, since γ reads the intervening tokens, so it is not a clean position channel. And it falls out of the recurrence rather than being a decision to factor attention. **The two-attention-matrix construction has no counterpart in that literature**, which is consistent with the reading that MapEM’s AND-gate is the genuinely distinct piece of the MapFormer paper.

3.2 Where MapEM and TEM-t sit

Two mechanisms in this project’s own experiments were left out of the map. Both turn out to be placeable, and the placements are opposite.

MapEM does not leave the frame; it lifts it. An earlier version of the table above put MapEM under “breaks the single bilinear form”. That is wrong. With $(A_X)_{ts} = q_t^x \cdot k_s^x$ and $(A_P)_{ts} = q_t^p \cdot k_s^p$,

$$(A_X \odot A_P)_{ts} = (q_t^x \otimes q_t^p) \cdot (k_s^x \otimes k_s^p),$$

so the Hadamard product *is* a single bilinear form — in the tensor product of the content and position spaces, of dimension $d_x d_p$ rather than d (verified to 1.9×10^{-6} in float32). MapEM is therefore in the frame, evaluated on a lifted space: everything in §3.3 applies to it, with c and G

taking values in $\mathbb{C}^{d_x d_p}$ and the position factor entering every one of the $d_x d_p$ channels rather than being summed alongside content in d of them.

That is a sharper statement than “an AND-gate instead of an OR-gate”, and it says what the lift costs: a rank-one tensor $q^x \otimes q^p$ cannot represent a query that is content-*or*-position, only content-*and*-position, and there is no fallback channel when either factor is uninformative. It also predicts nothing about *when* EM wins — the regime narrative built on the AND-gate was retracted in this project on a pre-registered test, and the algebra here is deliberately silent about it. What the algebra does explain is the observed sensitivity to how the position channel is initialised: on Match-Query, a single p_0 beats the paper-faithful separate q_0^p, k_0^p by +0.358 (3/3 seeds, the separate form collapsing to 0.107 on one). *Withdrawn (audit 2026-09-10)*: the reason given here — “with no additive fallback a mis-set position factor multiplies the whole score rather than being outvoted” — read MapWM as additive, which it is not, and the sign of the effect depends on the task (on k -back recency the separate form is ahead, +0.128 at $n = 24$).

TEM-t is MapFormer with the two structural conditions removed — and it came first.

TEM-t (Whittington et al., arXiv:2112.04035) generates position recurrently, $e_t = \sigma(e_{t-1} W_{a_t})$ with σ a ReLU and W_a a general per-action matrix, and then sets $Q = K = W_e e$ and $V = W_x x$. Content is absent from the logit entirely; it enters only through the values.

Its logit is $e_t^\top W_e^\top W_e e_s$. Writing $e_s = W_{t \rightarrow s} e_t$ and $e_t = W_{0 \rightarrow t} e_0$, with W_e orthogonal this is $e_0^\top W_{0 \rightarrow t}^\top W_{t \rightarrow s} W_{0 \rightarrow t} e_0$ — the interval operator *conjugated by the prefix*. Conjugation is trivial exactly when the group is abelian. So:

transition group	logit depends on the interval alone?
abelian (block rotations), no σ	yes
general $SO(d)$, no σ	no
abelian, with ReLU	no
general $SO(d)$, with ReLU	no

(`verify_theory.py`; the abelian linear row agrees to 10^{-6} and the others differ in the first digit.)

MapFormer is precisely the surviving row: $SO(2)^{n_b}$ is abelian by construction and there is no nonlinearity on the path. TEM-t breaks both conditions, and either break alone is enough.

This is the mechanical explanation of a cost this project measured without explaining: no group law means no S_t , hence no parallel prefix scan, hence a sequential loop. At $L = 2048$, **TEMFaithful** is $120\times$ slower than at $L = 128$ and $1632\times$ slower than **Vanilla** in absolute terms — with $20\times$ *fewer* parameters (20,705 against 405,472). The constant includes Python-loop overhead, but the *shape* is the missing group law, not the implementation.

The historical order is worth stating plainly: TEM-t is 2022 and MapFormer is 2025, so on this axis MapFormer is the later paper, and what it contributes is the *restriction* — abelian, linear — that buys the parallel scan and the relative law. That is the same move as §2.6’s $K = 1$ argument, applied to the transition group instead of the convolution width.

Is PoPE a variant of Selective RoPE? No, and not narrowly. Selective RoPE *rotates* $\tilde{q}, \tilde{k}$; rotation is orthogonal, so it preserves $\mu = |\tilde{q}^{(c)}|$ and merely shifts $\psi = \arg \tilde{q}^{(c)}$. It therefore cannot reach PoPE, which needs three things outside that reach:

- $\mu = \text{softplus}(Q_{t,c})$, a strictly positive nonlinearity of a linear functional, against $\mu = \|(q_{2c}, q_{2c+1})\|$, the norm of a linear map — different function classes, neither containing the other;

- $\psi \equiv 0$. A rotation by $\theta_t = -\psi_t$ would impose it, but no rotation *from this family* can: Selective RoPE’s θ is an accumulated function of $x_{1..t}$ drawn from (6), while $-\arg(W_Q x_t)$ is an atan2 of two linear functionals of the *current* token alone, which that class does not contain;
- $n_b = d_{\text{head}}$ rather than $d_{\text{head}}/2$, which is a change of blocking, not of parameters.

Nor does the reverse inclusion hold: PoPE’s *content* factor is positive real before the positional phase ($\mu > 0$ and $\psi \equiv 0$), which no choice of Selective RoPE’s generator imposes. (The composite $\tilde{q}_t^{(c)} = \text{softplus}(Q_{t,c})e^{it\vartheta_c}$ of course goes negative whenever $\cos(t\vartheta_c) < 0$; the constraint is on the factor, not the product.) **The two are not nested in either direction.**

So the design space is a product, not a chain. Inclusion holds *within* the phase column and nowhere else:

phase generator	standard (μ, ψ)	PoPE $(\mu, \psi=0)$
RoPE	RoPE	PoPE-Flat
$\cap$ MapFormer	MapFormer	MapPoPE
$\cap$ Selective RoPE	Selective RoPE	(<i>unbuilt</i>)

The vertical inclusions are the exact ones of §2.4; the columns are a separate choice, and §4 is the statement that they compose because they touch disjoint factors of (5). The bottom-right cell is well defined and has not been built — and there is presently no reason to, since Selective RoPE’s generator does not beat MapFormer’s on either task here (§10).

Writing the unbuilt cell down. Substituting Selective RoPE’s angle into PoPE’s bilinear form is immediate, since they touch disjoint coordinates:

$$a_{ts} = \frac{1}{\sqrt{d_{\text{head}}}} \sum_c \text{softplus}(Q_{t,c}) \text{softplus}(K_{s,c}) \cos\left(\theta_s^{(c)} - \theta_t^{(c)} + \delta_c\right), \quad (23)$$

$$\theta_t^{(c)} = \tau \sum_{u \leq t} [\sigma(W_g x_u) \odot (\kappa * W_\omega x)_u]^{(c)}.$$

The only thing that does not carry over for free is the *width*. PoPE indexes one frequency per element, so c runs to d_{head} rather than $d_{\text{head}}/2$, and the whole generator must be rebuilt at twice the channel count. Measured on the implementation:

angle path	rotary width ($n_b = d_{\text{head}}/2$)	PoPE width ($n_b = d_{\text{head}}$)
MapFormer generator	448	768 (incl. δ)
Selective RoPE generator	16,833	33,793 (incl. δ)

So MAPPOPE spends 768 parameters on its angle, 0.38% of the model, while (23) would spend 33,793 — 44× more, and 14.2% of the whole model. Against a generator that does not beat MapFormer’s on either task measured here, at a rank knob that buys the same thing for 384.

Why PoPE composes with an accumulated phase and is inert with an index one. The two things PoPE does are *not independent choices*: deleting ψ is what *buys* the doubled frequency count, and the constraint is forced by geometry. A rotation acts on a 2-plane and has one angle, so d_{head} coordinates admit exactly $d_{\text{head}}/2$ rotary frequencies; there is no rotary scheme with d_{head} of them. PoPE escapes that ceiling only by not rotating at all — it builds $\mu_c e^{i\varphi_c}$ elementwise and

assembles the bilinear form directly, which costs the per-element phase and returns a per-element frequency. Same d_{head} real degrees of freedom of content either way, reallocated.

That makes the composition result mechanical rather than fortunate. Doubling the frequency resolution is worth nothing when the phase is an index clock read over a window shorter than its shortest period — which is why PoPE alone sits on the floor (0.509 against 0.506). *Corrected 2026-09-25: that was a 16-epoch recipe. Converged, PoPE with an index phase reaches 0.679 against index RoPE’s 0.805 at the training length and beats it beyond (0.639 / 0.607 against 0.470 / 0.374 at $T = 512/1024$), so “inert” is withdrawn (PAPER2X2_RESULTS.md).* It is worth something when the phase encodes a *wrapped, multi-dimensional* quantity that must be decoded jointly across frequencies, which is exactly the residue problem of §2.5. Hence 0.994 with path integration, and hence a prediction: **PoPE’s gain should scale with how much wrapping the phase actually does** — with map size and with sequence length — and should vanish on a task whose phase never wraps.

Tested, and half of it is refuted (POPE.WRAPPING.md, $n = 8$, grids $\{16, 32, 64, 128\}$, one batch). The *length* half holds 3/3. The *octave* half does not: the loss-matched gain is $+0.077/+0.095/+0.079$ across grid size — flat, not rising with $\log_2 N$. So PoPE’s gain is not buying relief from wrapping in the way this paragraph argues, and the argument above should be read as the motivation for a measurement that came back against it.

One asymmetry is worth naming, because it is the reason the two axes behave differently. Descending the phase column **removes** constraints and gains expressivity. PoPE **adds** one: it gives up the intrinsic phase in exchange for twice the frequencies, at an unchanged content budget of d_{head} real degrees of freedom. And in Selective RoPE’s own taxonomy PoPE is neither of their two selectivities — it is not a rotation and it is not a decay on the state transition, but a restriction on the read and write vectors themselves. It does not fit their frame any more than it fits CoPE’s.

3.3 One equation: the complex log-polar accumulator

PoPE and Selective RoPE are not nested, but they are not unrelated either — they constrain different coordinates of a single object, and writing that object down subsumes every mechanism in this note.

Work in the complex domain of §1 and put each token’s contribution in *log-polar* form. Let $c^q, c^k, G : \mathbb{R}^d \rightarrow \mathbb{C}^{n_b}$ and set

$$\boxed{\tilde{q}_t = \exp\left(c^q(x_t) - \overline{S_t}\right), \quad \tilde{k}_s = \exp\left(c^k(x_s) + S_s\right), \quad S_t = \sum_{u \leq t} G(x_u),} \quad (24)$$

componentwise, with $a_{ts} = \frac{1}{\sqrt{d_{\text{head}}}} \sum_c \text{Re} \left[\overline{\tilde{q}_t^{(c)}} \tilde{k}_s^{(c)} \right]$ as before. The conjugate on S_t is load-bearing and is the whole content of the sign discussion below: it negates the accumulated *magnitude* on the query while leaving the accumulated *phase* alone, and the Hermitian form’s own conjugation then turns both into differences over $t < u \leq s$. Verified numerically at 1.3×10^{-14} against the closed form; writing $-S_t$ instead takes the same error to 3.6×10^1 — the identity does not merely degrade, it fails, because the phase terms add rather than cancel (`verify_theory.py`). Writing $c = \gamma + i\pi$ and $G = \delta + i\eta$ with all four real:

$$a_{ts} = \frac{1}{\sqrt{d_{\text{head}}}} \sum_c e^{\underbrace{\gamma_{t,c}^q + \gamma_{s,c}^k + \sum_{t < u \leq s} \delta_{u,c}}_{\text{magnitude}}} \cos \left(\underbrace{\pi_{s,c}^k - \pi_{t,c}^q + \sum_{t < u \leq s} \eta_{u,c}}_{\text{phase}} \right). \quad (25)$$

(The query and key maps must be kept separate. An earlier draft wrote a single c for both, which forces $\tilde{k}_t/\tilde{q}_t = e^{2\text{Re} S_t}$, a positive real fixed by the accumulator — untrue of any attention with

$W_Q \neq W_K$, and inconsistent with the four objects $\gamma^q, \gamma^k, \pi^q, \pi^k$ the next display already uses. It also left PoPE’s key-side bias δ_c with no home in the table below.) **c is the instantaneous complex log, G the accumulated one.** Real parts are magnitude, imaginary parts are phase — which is Selective RoPE’s “real parts forget, imaginary parts encode position” read off the algebra rather than from diagonal SSMs.

Every mechanism is a constraint on (c, G) . The parameterisation is fully general — $c = \log$ of whatever linear map produces $\tilde{q}$ — so this is a change of coordinates, not a new model.

	Re c <i>inst. magnitude</i>	Im c <i>inst. phase</i>	Re G <i>acc. magnitude</i>	Im G <i>acc. phase</i>
RoPE	free	free	0	ω (const)
MapFormer	free	free	0	$\omega W_{\text{out}} W_{\text{in}} x$
Selective RoPE	free	free	0	$\tau \sigma(W_g x) \odot (\kappa * W_\omega x)$
PoPE	$\log \text{softplus}(Q)$	0	0	ϑ (const)
forget gate / GLA	free	free	$\log \gamma(x)$	0
MapPoPE	$\log \text{softplus}(Q)$	0	0	$\omega W_{\text{out}} W_{\text{in}} x$

This is the unification asked for. PoPE constrains c ; Selective RoPE and MapFormer constrain Im G . Disjoint coordinates, so neither contains the other (§3.1) and both can be imposed at once (§4) — the composition is not a design decision but an immediate consequence of which entries of one table each mechanism writes to.

Why the two accumulators carry opposite signs, and what that buys. The Hermitian form $\tilde{q}\tilde{k}$ *subtracts* phases and *adds* log-magnitudes. So an accumulator yields a *relative* quantity only if it is signed to match:

- Im G accumulated with the *same* sign on q and k gives $\sum_{t < u \leq s} \eta_u$ — relative by construction. This is the relative-position property of §1, and it is why a rotation gets it for free.
- Re G accumulated with the same sign would give $\sum_{u \leq t} + \sum_{u \leq s}$, a joint scale carrying no information about separation. Signing it *oppositely*, as (24) does, gives $\sum_{t < u \leq s} \delta_u$ — and with $\delta = \log \gamma < 0$ that is exactly the forget gate’s $\prod_{t < u \leq s} \gamma_u$.

Measured by perturbing the *prefix* $u \leq t$ only and asking whether the score moves — a relative score must not (`verify_theory.py`):

Re G	Im G	relative prefix leak
opposed	aligned	3.4×10^{-16}
opposed	opposed	1.4
aligned	aligned	3.1
aligned	opposed	7.1

Exactly one of the four combinations is interval-relative, and it is the one (24) writes. An earlier draft of this table had the two rows backwards; so did the first version of the script that checks it, which parameterised its two branches inconsistently and had to be rewritten against $\tilde{q}, \tilde{k}$ as written. So the forget gate is not merely the magnitude analogue of path integration; it is the *antisymmetrically* accumulated one, and the antisymmetry is what turns a scale into a decay over distance. A symmetrically accumulated magnitude is a well-defined object that no mechanism here uses, and it would encode “how much total stuff has happened by t and by s ” rather than anything relative.

The space of relative accumulators is CLOSED, and both slots are taken. With content-dependent increments, “relative” cannot mean “a function of $s - t$ ” — it means *a function of the interval $(t, s]$ alone*, i.e. insensitive to tokens before the query. Testing that directly, by perturbing a token outside the interval and measuring the change in the logit:

coordinate	q, k exponent	change	code	occupied by
$\text{Im } G$	aligned	1×10^{-13}	relative	RoPE, MapFormer, Selective RoPE
$\text{Im } G$	opposed	1.8×10^2	leaks the prefix	—
$\text{Re } G$	opposed	4×10^{-13}	relative	forget gate, GLA, Mamba, FoX
$\text{Re } G$	aligned	1×10^8	leaks the prefix	—

Given the parameterisation, the two unoccupied rows are excluded: exactly one sign choice per coordinate yields a relative code, and the literature occupies both.

CORRECTION — “there is no third slot to invent” is **FALSE** as printed, and the counterexample lives inside the commuting family. The closure argument silently assumes the per-channel transition is *semisimple*: that the interval acts by a scalar in $\mathbb{C}^\times \cong \mathbb{R}_{>0} \times S^1$, whose only two coordinates are Re log and Im log. Commuting is weaker than diagonalisable. Over $\mathbb{R}$ there are exactly *three* unital commutative 2-dimensional algebras — $\mathbb{C}$, $\mathbb{R} \oplus \mathbb{R}$ (a degenerate double copy of the magnitude slot), and the **dual numbers** $\mathbb{R}[\varepsilon]/(\varepsilon^2)$, whose group is the unipotent shear $\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$.

The shear slot is real and I have checked it: it commutes and satisfies the group law exactly; it is interval-invariant to 1.5×10^{-16} (`verify_theory.py`) under the perturbation test above; and it contributes

$$q^\top M_t^{-1} M_s k = q_1 k_1 + q_2 k_2 + \left(\sum_{t < u \leq s} b_u \right) q_1 k_2,$$

a term **linear in the accumulated interval quantity** with a content-dependent coefficient. Since $e^{\text{Re}} \cos(\text{Im})$ is an exponential times a sinusoid, it can never be polynomial in the accumulator, so this is outside (24) while remaining abelian, per-token, and parallel-scannable. The kernels it generates are *polynomial* in the accumulated interval quantity — linear here, optionally times a real exponential, i.e. $e^{-\gamma d}(\alpha + \beta d)$ rather than $e^{-\gamma d} \cos(\omega d)$.

It is also *occupied*: Jordan-RoPE (arXiv:2605.04217) builds the defective block directly, and GRAPE (arXiv:2512.07805) uses the same nilpotent algebra as its “additive” family, recovering ALiBi and the Forgetting Transformer from it.

The claim that survives. *For per-token increments valued in a fixed maximal torus of $GL(d)$ — simultaneously diagonalisable, semisimple transitions — the relative accumulators are exactly $\text{Re } G$ and $\text{Im } G$, and both are occupied.* That is a real result and it is the precise sense in which commutativity buys the parallel scan at the cost of the non-abelian *and* non-semisimple design space. Mechanisms that appear to add a slot break one of four things rather than adding a scalar: the *fixed basis* (PaTH, DeltaNet, DeltaProduct, RWKV-7 — the interval acts by a path-ordered product, so no accumulated S_t exists); *per-token factorisation* (CoPE — position becomes query-dependent); *the path structure itself* (MesaNet, log-linear attention, Titans, and TAPE, which updates its positional tensor across layers through attention and MLP rather than by accumulating per-token increments); or *semisimplicity* (the shear slot above).

That is a closure result rather than an opening one, and it is the more useful kind. It says any new relative-position mechanism in this family must be a different (m, φ, K) *inside* one of the two existing slots, not a new slot — which retro-explains why RoPE, its frequency-schedule variants (YaRN, LongRoPE, FoPE) and Selective RoPE all cluster where they do. *Corrected*: an earlier version of this sentence also listed PaTH and CoPE. Neither belongs — PaTH is non-abelian and

provably outside (its Theorem 2.1 puts a one-layer two-head PaTH transformer on an NC^1 -complete problem, i.e. outside TC^0 , on their stated assumption $TC^0 \neq NC^1$, where every mechanism inside this frame sits), and CoPE is excluded two sections later for a different reason again. And it sharpens what MapFormer is missing: not “some unexplored cell”, but the *only other slot there is*.

Why the two slots are parameterised so differently. The coordinates are bounded in different ways, and their parameterisations follow. The phase is *periodic*, so an accumulated angle is ambiguous modulo 2π and must be decoded jointly across scales — hence 32–64 frequency blocks, hence the whole rank and residue argument of §2.5. The magnitude is *unbounded below* and merely underflows, so a single rate suffices — hence the forget gate is one scalar per head (259 parameters here against the phase path’s 448–33,793). **Prediction: a multi-frequency forget gate buys nothing**, while a single-frequency phase is crippled. Neither has been run, and the asymmetry is cheap to test in both directions.

What the table says is missing. MapFormer, Selective RoPE and PoPE between them leave two entries untouched *in the published rows*. $\text{Re } G$ is 0 in every row of the table — but this project has since built and measured it: `model_forget.py` adds $\log \gamma_t = -\lambda \sigma(W_f x_t)$ for 259 parameters and buys +0.086 loss-matched at $T = 1024$, 8/8 seeds, at $r = 2$ (`FORGET_GATE.md`). The mechanism is *not* identified: the gain is anti-correlated with λ ($r = -0.516$), 5/8 seeds learn $\lambda < 0$, and the frozen- λ control lands on Vanilla, so it needs a live λ but not the decay. And $\text{Im } c$, the instantaneous phase, is only ever *free* or *deleted*; nothing designs it. The second may be principled: c is instantaneous, so a constraint on it removes degrees of freedom without creating a code, whereas a constraint on G propagates over the whole sequence. That would predict that constraints pay on the accumulated coordinates and not the instantaneous ones — consistent with low rank helping on $\text{Im } G$ (§2.5) (PoPE’s apparent inertness alone was a 16-epoch artefact, §4), but it has not been tested as a claim in its own right.

4 MapPoPE: the conjunction

Since MapFormer determines φ and PoPE determines μ , combining them requires no new mechanism — only substituting one into the other. Replace PoPE’s positional phase $t\vartheta_c$ with the path-integrated angle (9):

$$a_{ts} = \frac{1}{\sqrt{d_{\text{head}}}} \sum_c \underbrace{\text{softplus}(Q_{t,c}) \text{softplus}(K_{s,c})}_{\text{PoPE: magnitude from content}} \cos \left(\underbrace{\theta_s^{(c)} - \theta_t^{(c)}}_{\text{MapFormer: path-integrated phase}} + \underbrace{\delta_c}_{\text{PoPE bias}} \right), \quad (26)$$

with $\theta_t^{(c)} = \omega^{(c)} \sum_{u \leq t} \Delta_u^{(c)}$ as in (9). In code the class is three lines: inherit MapFormer’s forward pass (so $\cos \theta, \sin \theta$ still come from the path integrator), widen the angle machinery, and swap the attention module.

The one non-trivial detail: frequency count. RoPE rotates *pairs* of coordinates, so it needs $n_b = d_{\text{head}}/2$ angles. PoPE assigns one frequency per *element*, so it needs $n_b = d_{\text{head}}$. The combination therefore rebuilds both position modules at the wider count:

$$n_b : d_{\text{head}}/2 \mapsto d_{\text{head}}, \quad W_{\text{out}} \in \mathbb{R}^{(H d_{\text{head}}) \times r}, \quad \omega \in \mathbb{R}^{H \times d_{\text{head}}}. \quad (27)$$

In parameters this is small: at $d_{\text{model}} = 128$, $H = 2$, one layer, the two models are **204,373** and **204,693**, a difference of +320 (+0.16%) — +64 on ω , +128 on W_{out} , +128 for δ_c . The consequential change is not the parameter count but the **frequency count**: MAPPOPE carries twice as many independent position frequencies, indexed per element rather than per pair. So the two are neither exactly parameter-matched nor frequency-matched, and the second is what a comparison has to contend with.

Why the parameter count barely moves: the content budget is conserved. The widening is not a capacity increase, and the arithmetic says why. Ordinary rotary attention groups d_{head} coordinates into $d_{\text{head}}/2$ complex numbers, each carrying a magnitude *and* an intrinsic phase — d_{head} real degrees of freedom of content per head. PoPE assigns one frequency to each of the d_{head} elements and keeps only a magnitude — again d_{head} . **PoPE spends the same content budget differently**: it buys twice the frequency resolution by giving up the per-block phase ψ . That is the same trade as the “restriction” noted in §3, counted rather than described, and it is the reason the two models differ by 320 parameters rather than by a factor.

What this does and does not contaminate. In the $2 \times 2 \{\text{RoPE}, \text{PoPE}\} \times \{\text{index, path-integrated}\}$, both PoPE cells are wide and both RoPE cells narrow, so the width confound is *constant along the encoding factor*. The position main effect and the interaction are therefore clean; only the encoding main effect carries it — and under the 16-epoch recipe that was the small one (+0.003 on the torus against +0.461 for position, both paired $n=8$ main effects on the held-out map). Under the converged recipe it is not: $-0.049 / +0.114 / +0.189$ at $T = 128/512/1024$, all detectable, against position +0.243 at $T = 128$ (PAPER2X2_RESULTS.md), so the width confound bears on a measurable effect.² There is also a direct empirical check: if the extra frequencies were doing the work, PoPE with an index phase should beat RoPE with an index phase. It does not (0.509 vs 0.530, both on the 0.506 floor, 16-epoch recipe; converged, 0.679 against 0.805 at the training length, though PoPE leads beyond it). The missing control is a *narrow* PoPE at $n_b = d_{\text{head}}/2$ with each angle repeated across its pair, which would isolate the parameterisation from the width; it has not been run.

The widening is a property of PoPE itself, not of the combination — the index-phase PoPE baseline also runs at $n_b = d_{\text{head}}$.

4.1 Why the two compose, and when they do not

(26) predicts the observed pattern.

- **PoPE alone is inert where position carries the signal — WITHDRAWN.** Under the 16-epoch recipe PoPE with an index phase scored 0.509 against a floor of 0.506. Converged it reaches 0.679, below index RoPE’s 0.805 at the training length and above it past it (0.607 against 0.374 at $T = 1024$): not inert, and the sign depends on the length.
- **PoPE plus path integration is the strongest configuration measured.** 0.994 on the same task. Once the phase carries genuine position, sharpening the content channel pays.
- **On text the picture does not survive its own metric.** Reading the *final* checkpoint gives PoPE alone -0.0053 bpc, path integration alone -0.0041 , sum -0.0093 , both together -0.0058 — apparently sub-additive. But the within-run checkpoint standard deviation is 0.003–0.007,

²CORRECTED 2026-09-06. BASELINE_TABLE.md, CLAUDE.md and MINIGRID_2X2X2.md all printed +0.011 in this cell, with a “40×” ratio derived from it. That +0.011 is MapPoPE-Flat – MapWM-Flat, the path-integrated *row* alone at $n=3$ — not the encoding main effect — and the ratio inherited the mismatch. All three files now carry the $n=8$ figure.

larger than every one of those quantities, which is why the source file specifies the **mean of the last five checkpoints** instead. On that metric the numbers are -0.0041 , -0.0006 , sum -0.0047 , both -0.0052 : path integration alone nearly vanishes and the combination is mildly *super-additive* — the opposite sign. Both single-axis arms are also $n = 1$. **Nothing about additivity on text is established here in either direction**; the two readings are reported so the fragility is visible rather than resolved by choosing one.

The general statement: the two mechanisms are independent as *parameterisations* — they touch disjoint factors of (5) — but that does not make their *effects* independent. Whether they add depends on whether the task has separate work for a phase and a magnitude to do.

4.2 The classical relative methods, and a fourth way out of the frame

Shaw-style relative position representations, Transformer-XL and T5’s relative bias predate the rotary line and are still widely deployed. They belong in the map, and they occupy a slot the out-of-frame table did not have.

Shaw et al. add a learned vector indexed by the *clipped lag*,

$$s_{ij} = \frac{q_i^\top (k_j + a_{r_{ij}}^K)}{\sqrt{d_{\text{head}}}}, \quad r_{ij} = \text{clip}(j - i, -K, K),$$

and T5 compresses the same idea to a per-head scalar, $s_{ij} += b_{h, \text{bucket}(j-i)}$. Transformer-XL carries the relative form across cached segments.

These are linear in q and k and translation-invariant, but the map from lag to operator is a **lookup table over a discretised lag**. That fails Puranik’s third condition, *continuity*, and with it the group law: there is no generator M because clip and bucket are step functions, and beyond the table’s range the operator is constant rather than continuing. So they leave the frame by a route distinct from non-abelian transport, pair-dependence or a broken path structure.

The trade-off is exact and worth stating, because it is the one design choice the rotary line silently makes. **A table can represent an arbitrary function of the lag inside its window and cannot extrapolate outside it at all; a one-parameter group is restricted to the three families of §6 and extrapolates by construction.** RoPE gives up expressiveness over short lags to buy a functional form that is defined at every lag. Whether that purchase is honoured is a separate question, and the long-context literature’s own conclusion is that it frequently is not: the ability to *compute* positional features beyond the training length does not imply reliable generalisation there, which is why context extension is evaluated by short-context retention, position-wise perplexity, retrieval and long-context reasoning rather than by whether the encoding is defined.

5 What leaves the frame

Two kinds of mechanism are outside (5), and it is worth being explicit about which, since both are routinely grouped with the ones above.

CoPE, and why it is not in this family. Contextual Position Encoding computes a gate from a *query-key pair*, $g_{ts} = \sigma(q_t \cdot k_s)$, and takes the position of s relative to t to be $\sum_{s < i \leq t} g_{ti}$. That quantity depends on both indices and on the query, so it cannot be written as $\theta_s - \theta_t$ for any per-token θ : the factorisation that (4) relies on — and with it the relative-position property that holds for *any* θ — does not exist. **CoPE is not a phase generator with a different (m, φ, K) ;**

it leaves the frame. This is worth stating because CoPE, Selective RoPE and MapFormer are often listed together as “content-dependent position”, and only the last two are the same machine.

PoPE is not in the CoPE family either, despite the name. Its magnitude is content-dependent; its *angle* is not, apart from the constant bias δ_c (§3). It modifies the other factor.

Mechanisms that act elsewhere in the network. Weight-shared recursion, hierarchical pooling and the InEKF correction are not choices about μ or φ at all: the first two change how the residual stream is processed at fixed θ , and the third post-processes θ before the rotation. Appendix A gives all three in this notation, together with what each was measured to be worth — which in the InEKF’s case is a withdrawal.

6 Prior art: most of this frame is published

Read after the fact, and recorded here because it changes what any of the above can be claimed as.

The frame is a published classification, and it has a proof. Alok Puranik, *Using group theory to explore the space of positional encodings for attention* (Jane Street Blog, 22 April 2026), derives §3.3 from three assumptions and gets a stronger result than this note states. Take $\tilde{q}_t = F(t)q_t$, $\tilde{k}_s = G(s)k_s$ so the logit is $q_t^\top F(t)^\top G(s)k_s$; require *linearity*, *translation invariance* ($F(t)^\top G(s)$ depends only on $s - t$), and *continuity*. Normalise so that $F(t)^\top G(t) = I$ (absorbing the constant $A(0)$, which the three assumptions alone do not force, into the keys). Then $A(d) := F(0)^\top G(d)$ satisfies $A(0) = I$ and $A(d)A(e) = A(d + e)$ — a one-parameter matrix group — so $A(d) = \exp(dM)$ for a fixed generator M , and the classification of positional encodings *is* the classification of M up to conjugacy:

- M diagonalisable, real eigenvalue λ : $e^{\lambda d}$, i.e. exponential decay ($\lambda < 0$), NoPE ($\lambda = 0$), or a blow-up that is discarded.
- M diagonalisable, conjugate pair: real canonical form $e^{\lambda d} \cdot \text{rot}(\omega d)$ — damped RoPE. **This is (24)’s Re / Im split, and the reason there are exactly two slots is that $\mathbb{C}$ has exactly two.**
- M defective: polynomial factors d^r . He reports finding no literature on defective generators for positional encoding and calls the interpretation unclear; ALiBi is then shown to be realisable this way after augmenting q, k .

He also states the reparameterised-time reading directly: gated models with a data-dependent decay should be understood as “learning how far to advance time, as opposed to changing the rate of decay” — which is $S_t = \sum_u G(x_u)$ replacing t , published as an aside. **Jordan-RoPE** (arXiv:2605.04217, May 2026) then fills his defective cell: a complex eigenvalue coupled to a nilpotent *in the same block*, giving modes $d^r e^{i\omega d}$, and — because such blocks are not orthogonal — requiring a *contragredient* query action to recover a pure relative score. That last point is the q/k sign asymmetry of §3.3 in its non-semisimple form; it is also published.

GRAPE (Group Representational Position Encoding; Zhang, Chen, Liu, Qin, Yuan, Xu, Yuan, Gu, Yao; arXiv:2512.07805, ICLR 2026) is this note’s two-slot decomposition, published. Verified against the paper itself, not a summary:

- **Both slots.** Its abstract: “GRAPE brings together two families of mechanisms: (i) multiplicative rotations (Multiplicative GRAPE) in $SO(d)$ and (ii) additive logit biases (Additive GRAPE) arising from unipotent actions in the general linear group GL ”. On the special cases, the sentence that says it is “This perspective recovers RoPE and ALiBi as exact special cases, proves that FoX is an exact instance of Additive GRAPE” — an earlier draft spliced in the abstract’s “as exact special cases while preserving an exact relative law”, whose subjects there are ALiBi and FoX, not RoPE.
- **The “third slot” of §3.3 is GRAPE’s additive family.** Their additive side *is* the unipotent/nilpotent algebra — the shear whose omission from (24) that section corrects. The correction is right and it is not new.
- **The accumulated-magnitude slot, with a content-dependent increment.** They prove FoX is an exact instance: “Let $f_t \in (0, 1]$ be per-token forget scalars and set $\omega_t := \log f_t$. The resulting additive bias is $b(t, j) = \sum_{\ell=j+1}^t \omega_\ell$, which coincides with FoX’s forgetting bias D_{ij} .” That is exactly $\text{Re } G = \log \gamma(x)$ accumulated over the interval.
- **And a content-dependent accumulated phase.** Path-Integral Additive GRAPE: the bias depends on “a contextual phase difference $\Phi_j - \Phi_i$ when using cumulative phases $\dots \Phi_t$ is a scalar path variable, typically defined as a cumulative sum $\Phi_t = \sum_{\ell < t} \omega_\ell$ of per-token frequencies ω_ℓ , so that $\Phi_j - \Phi_i$ plays the role of an effective relative position.”

What that leaves. Not the taxonomy, and not the closure argument. On what I read, GRAPE’s content-dependent cumulative phase enters an *additive* linear-in-offset bias on the unipotent side, while its *multiplicative* angle is indexed by n and its extensions there concern generator geometry (learned commuting subspaces, compact non-commuting mixtures) rather than content-dependence of the angle.

Boundary, now settled — Appendices C, D, E and J read. It holds, and more sharply than expected.

- **App. C** confirms the magnitude slot exactly: prefix sums $U_{t,h} = \sum_{\ell < t} \log f_{\ell,h}$, bias $D_{ij,h} = U_{i,h} - U_{j,h}$, “FoX is an exact GRAPE-A / GRAPE-AP instance”. This is $\text{Re } G$ with a content-dependent increment, published and proved.
- **App. D** is about the *spectral* structure of the non-commuting multiplicative extension, and labels it in as many words “the expressive power of **non-contextual** non-commuting MS-GRAPE”. The multiplicative side is not contextual.
- **App. E** composes the two as one block-diagonal one-parameter subgroup $G_{\text{joint}}(m) = \exp(mL)$, $L = L_{so(d)} \oplus \omega \Lambda A_0$ — and it is indexed by the *offset* $m = j - i$. The content modulation $\Lambda = \lambda_q + \lambda_k$ enters only the nilpotent block.
- **App. J.4** shows GRAPE is aware of contextual multiplicative maps, but only through PaTH, which it critiques as contractive and near-singular — and PaTH is non-orthogonal and non-commuting.

So GRAPE’s content-dependence lives entirely on the additive side, and there it carries a constraint worth noting: the phase-modulated form requires $\omega_\ell = g(x_\ell) \geq 0$, i.e. a *monotone* clock, whereas MapFormer’s Δ is a signed linear map and can represent a -1 action.

Scoped on audit. “Not contextual” is App. D’s word for one sub-case (the thin compression $L = EL_r E^\top$), not for GRAPE-M as a family. The paper claims contextual multiplicative variants

in at least four places — Fig. 1’s caption (“also extends to contextual forms via state-dependent generators”), the related work (“phase-modulated and dictionary-based contextual variants ... cumulative token-adaptive phases”), App. F (“the GRAPE-M scale cleanly translates to contextual versions where frequencies or phases depend on the token content”) and App. J.4 (“both non-contextual and contextual types”). What I could not find is any *formal definition* of one in the body. So the defensible claim is that GRAPE does not formalise a contextual multiplicative angle, not that its multiplicative side is non-contextual; the table’s GRAPE-M row (A3 = index) carries the same caveat.

What is left, stated precisely. GRAPE occupies *index*-multiplicative and *content*-additive. PaTH occupies *content*-multiplicative but non-abelian, at the cost of orthogonality and of any accumulated S_t . **Content-dependent, signed, orthogonal — i.e. contextual multiplicative and still abelian — is the hole between them**, and it is exactly where MapFormer, Selective RoPE and CARoPE sit. The three are not interchangeable inside it: CARoPE’s increment is monotone, and §10.2 measures what that costs on navigation — at matched training loss it gives up the whole of path integration’s advantage over an index clock. That is a narrow claim compared with “a unified theory of positional encoding”, and it is the one the literature actually leaves open.

Mamba-3 closes that hole. Verified first-hand. arXiv:2603.15569 (Lahoti, Li, Chen, Wang, Bick, Kolter, Dao, Gu; March 2026). Its complex SSM is $\dot{h} = \text{Diag}(A(t) + i\theta(t))h + (B(t) + i\hat{B}(t))x(t)$, with *both* A and θ functions of t . Proposition 2 discretises this to a real transition $e^{\Delta_t A_t} R_t$ with $R_t = \text{Block}\{R(\Delta_t \theta_t[i])\}$ — “a scalar decayed block-diagonal matrix of 2×2 **data-dependent rotation** matrices”. Proposition 3, titled *Complex SSM, Data-Dependent RoPE Equivalence*, shows the rotations accumulate as a product $\prod_{i=0}^t R_i$ applied to B and C — “which, by the state space duality framework, correspond to the query (Q) and key (K) components of attention”. Their own summary: “our complex-valued update rule is equivalent to a **data-dependent rotary embedding**”.

That is content-dependent, signed ($\theta(t) \in \mathbb{R}^{N/2}$, unconstrained, unlike GRAPE-AP’s $\omega \geq 0$), abelian, accumulated, applied to Q and K — every property of the cell named above. **The frame’s remaining theoretical gap is closed, by the Mamba authors, in March 2026.**

What is actually left, stated without inflation. Not the taxonomy, not the closure argument, not the content-dependent rotation. Two things survive, and both are empirical rather than structural:

- **The rank bottleneck — the *input-side* one.** Stated too broadly in an earlier draft (“none of them constrains the angle map to low rank”), which the same document falsifies two pages earlier: Selective RoPE uses “a low-rank MLP with rank 8 or 16” on the ω projection, and GRAPE’s abstract specifies “a rank-2 skew generator” and “rank-1 (or low-rank) unipotent actions”. What none of them constrains is the *input* side — how many dimensions of token content may steer the angle — as a designed variable swept against performance. Their ranks are stability and efficiency choices; MapFormer’s r is an inductive bias about the dimensionality of the action space. Mamba-3’s $\theta(t)$ is a per-channel projection with no such bottleneck. The measured result — $r = 2$ is the worst setting, $r = 4$ buys +0.085 at $8\times$ training length ($T = 1024$ against a training length of 128; at $4\times$ it is +0.038) for 384 parameters, 8/8 seeds, and trained at $T = 1024$ the rank per head decides whether training finds the solution at all (0/8 shared $r = 2$, 2/8 the paper’s per-head $r = 2$, 8/8 $r = 4$) while a rank-2 solution exists —

is a claim about a constraint that nobody else sweeps above the world dimension (MapFormer itself ablates $r \in \{1, 2\}$).

- **The navigation regime.** The papers above evaluate on language, recall, state tracking, and — for LieRE and Algebraic Positional Encodings — image, video and tree-transduction benchmarks. None runs grid navigation, and the one result this project has that survives every audit — that path integration is necessary for in-context cognitive maps, against a measured floor and a context-destruction ablation — lives there.

This is the more useful of the two, because it is where a difference the language side cannot see becomes decisive. §10.2 is the worked case: the *sign* of the increment is settled prior art on parity and formal languages (Sarrof, Grazi, and Selective RoPE’s own §4.2), and is invisible in the positional-encoding-for-language literature, where CARoPE, GRAPE-AP and CoPE are all monotone by squashing-function accident and none cites any of it. On navigation it costs *everything*: at matched training loss a monotone content-dependent phase does not beat a plain index clock at any length, while a signed one beats it by +0.123 and +0.195 at $4\times$ and $8\times$ training length on 12/12 seeds. A regime that makes a published-but-unapplied constraint measurable is worth more than a regime that merely adds a benchmark.

Verified 2026-09-06. Every source below was read first-hand and is stored under `papers/` (index: `papers/INDEX.md`). Three checks came back stronger than reported.

CARoPE (arXiv:2507.23083, July 2025) does accumulate a content-dependent angle, predating both Selective RoPE and MapFormer, and its increment is

$$f(x_t) = \frac{1}{\text{softplus}(x_t W) + 1} \in (0, 1), \quad \phi_i(m) = \sum_{t \leq m} (f(x_t)_h)^i,$$

so it is not merely non-negative: it is bounded strictly inside $(0, 1)$ and can neither reverse nor *stay*. It also differs on a second axis — $W \in \mathbb{R}^{d \times h}$ emits **one scalar per head**, so its content-to-angle map is rank 1 per head, *below* MapFormer’s r rather than above it. A monotone rank-one clock cannot represent a -1 action at all, which is the sharpest structural separation in the table. **Measured in §10.2:** on navigation, CARoPE’s own parameterisation reaches an opposition score of 1.98 out of a possible 2.00 — its east and west are all but the same vector — and at matched training loss the arm carrying it does not beat a plain index clock anywhere.

HGRN (arXiv:2311.04823) Table 11 ablates exactly this and concludes, in its own words, that “the phase argument θ should not be data-dependent”; HGRN2 (arXiv:2404.07904) Table 1 shows real HGRN at twice the state size beating complex HGRN, attributing the gain to state expansion rather than to the phase — though note this is evidence about *complex recurrence* at matched real capacity, and HGRN’s own θ is data-independent by design (“we find that parameterizing θ_t in a data-independent manner is preferable”), so it speaks to a neighbouring axis rather than to content-dependence itself. With our own (underpowered) enwik8 result, that is one direct negative and two adjacent ones, all on *language*. None of them evaluates navigation, which is precisely why the surviving claim is scoped to that regime.

LieRE (arXiv:2406.10322, ICML 2025) is the nearest published neighbour to the rank axis and must be cited as such. It learns a skew-symmetric basis $\{A_i\}$ and rotates by $R(p) = \exp(\sum_i p_i A_i)$ — structurally MapFormer’s **ActionToLieAlgebra**, driven by a *known position* instead of by content. Its Fig. 6 sweeps the generator’s tile size 2×2 through 48×48 and **peaks at 8×8** : an interior optimum, denser is worse. That is the same *shape* of finding as our step at $r = 2$, on a different axis — LieRE varies density on the output side (how much of $\mathfrak{so}(d)$ the rotation may occupy), r varies

the bottleneck on the input side (how many dimensions the content may steer). Neither subsumes the other, and a reviewer will ask.

7 The memory reading, and the readout axis

Two literatures work on the same equation from other directions. Both were absent from this note and both change what the map is a map of.

7.1 Fast weights, linear attention, and the primal/dual choice

The outer product $v_u \otimes k_u$ binds a value to a key; superposing bindings gives an associative memory read by a query. It appears as Kohonen’s correlation-matrix memory (1972), Schmidhuber’s fast weight programmer (1992), the recurrent form of linear attention (Katharopoulos et al., 2020), the hidden state of a scalar- A state-space model, Smolensky’s tensor-product role-filler binding (1990), and — via the dual form of Irie et al. (2022) — the weight matrix of any linearly-parameterised layer trained by gradient descent. Schlag et al. (2021) made the linear-attention correspondence explicit. Every mechanism therefore has two readings,

$$\underbrace{W_t = \sum_{u \leq t} A(t-u) v_u k_u^\top, y_t = W_t q_t}_{\text{primal: compress}} \iff \underbrace{a_{ts} = \tilde{q}_t \cdot \tilde{k}_s \text{ over kept items}}_{\text{dual: keep and attend}}$$

and $A(d)$ — the object §6 classifies — is the memory’s **transport kernel**: what happens to a stored trace as it ages d steps. $\text{Re } G$ is how fast it is forgotten; $\text{Im } G$ is how it is re-addressed. Three consequences.

A8 is not a separate axis. A positional mechanism admits a parallel prefix scan exactly when the memory admits a bounded recurrent state, because both are the group law $A(d)A(e) = A(d+e)$. Primal/dual and scannable/not are one condition.

Positional mechanisms are key-side by construction. Gershman, Fiete and Irie (2025) argue that key-value memory’s distinguishing feature is separate representations for storage and retrieval, and that the binding constraint on memory is addressing rather than capacity. Every row of the map acts on q and k and none touches v : **the frame is a theory of addressing**, and the two slots are the two ways an address can move — transported around a torus, or faded.

Why the magnitude slot is compulsory in the SSM rows and optional in the softmax rows. Schlag et al. show the additive outer-product memory has finite capacity and enters an overcapacity regime once the sequence exceeds it. In the primal form decay is capacity management — something must evict — while in the dual form items are kept separately and the softmax renormalises, so nothing saturates and decay (ALiBi, FoX) is an inductive bias one may decline. Same equation; the slot is mandatory on one side of the line and free on the other. It also reframes the delta rule: $I - \sigma(\beta_t) \phi \phi^\top$ is introduced because additive updates cannot *edit* a stale association, so rewritability is a capacity fix that brings state-tracking expressivity with it — and with $\sigma = 2$ sigmoid its eigenvalues lie in $(-1, 1)$ including negatives, which is §10.2’s axis exercised from the memory side.

The two literatures have divided the slots without noticing. Irie and Gershman’s review (TMLR 2025) categorises the entire modern linear family by *decay type* — RetNet a constant

scalar, Mamba-2 a scalar, GLA a rate per row — and the strings `RoPE`, `rotary` and `positional` occur **zero** times in it (checked case-sensitively; a case-insensitive grep for `rope` matches `property` and reported 17 false hits). The hybrid quadratic-linear transformers of Irie et al. (2025) use RoPE only as an off-the-shelf component inside the attention half. Mamba-3 claims both slots deliberately, arriving from the memory side; Selective RoPE had paired an input-dependent rotation with decay gates four months earlier (“recall needs rotation and decay”).

7.2 The readout, and the critical dimension

§2 treats the readout $A = \text{diag}(\omega)$ as fixed. The long-context literature varies exactly that: Position Interpolation, NTK-aware scaling, Dynamic NTK, NTK-by-parts, YaRN, LongRoPE and LongRoPE2 all rescale the frequency ladder. All are index-driven, so they occupy one cell of the map — but they vary something the eight axes do not name.

Their organising argument transposes usefully. Channel r has wavelength $T_r = 2\pi b^{2r/d_{\text{head}}}$; high-frequency channels complete many cycles inside the training length L_0 and are trained through their whole phase range, low-frequency channels may not complete one, and $r_{\text{crit}} = \min\{r : T_r \geq L_0\}$ separates them. Beyond L_0 the channels past r_{crit} are read at phases never seen in training — 36 of 128 dimensions for LLaMA-2.

Tested, and the account is wrong. Transposed to a content-dependent phase the coordinate is the accumulator rather than the index, which gives a prediction: the benefit of the OOD-helping mechanisms should localise to the channels whose period exceeds the trained range of S_t , and ablating those channels should cost *less* at extrapolated length because they are already being read out of distribution.

The precondition holds — at the training length 8–17 of 64 channels never complete a cycle of their own period, and at $8\times$ that length only 1–4 do — so the mechanism could operate. It does not. Ablating the 32 lowest-frequency channels is free at training length (-0.001) and costs 0.14–0.25 at $T = 1024$, on every arm whose baseline is off the floor. **Those channels are more load-bearing at extrapolated length, not less**, which is the opposite of the prediction. Damage is localised in frequency — high-frequency ablation costs 0.42–0.57 — but the plainer reading wins: the low-frequency channels carry position at map scale, and coarse position matters more when the walk covers more of the map.

What the same measurement did find: the accumulator’s growth law. All four arms start from the same accumulator range at training length (88–94) and separate entirely by how fast it grows with sequence length, $\text{range}(S) \sim T^\alpha$:

arm	$T=128$	$T=1024$	α	opposition score
signed, $r=4$	91.8	269.7	0.518	0.125
signed, $r=4$ (second batch)	94.1	279.7	0.524	0.092
signed, $r=2$	88.5	320.5	0.619	0.495
monotone, $r=4$	91.5	649.8	0.943	1.849

A signed, cancelling increment gives $\alpha \approx 0.52$: the accumulator performs a **diffusive random walk**. A monotone one gives $\alpha \approx 0.94$: **ballistic drift**. The last column is the opposition score $\|\Delta(+x) + \Delta(-x)\|/\|\Delta\|$, measured in a different batch for a different purpose; across the four arms $r(\text{opposition}, \alpha) = +0.9995$.

This is one observable for two separate results, not a mechanism. How completely opposite actions cancel sets whether the accumulator diffuses or drifts; that sets how fast it leaves

the range it was trained on; and that *co-varies* with the rate of degradation with length — the growth law is an association, not a cause (see “The growth law is an association” below). A rank-2 code (opposition 0.495, an arm mean) and a monotone increment (1.849) lie on one line, but within $r = 2$ opposition does not predict accuracy and the rank deficit is search (§2.5). Four points, correlational across arms; $\alpha \approx 0.5$ for a signed walk is expected in itself, and what carries the result is the spread to 0.94 and the cross-batch agreement with a quantity measured for another reason.

Scope of the existing surveys. The most recent survey of position encoding (Li, arXiv:2608.10021, August 2026) covers absolute and relative methods, RoPE and long-context scaling, and contains **zero** occurrences of CoPE, MapFormer, Selective RoPE, CARoPE, FoX, Mamba or GRAPE. The 2023 length-extrapolation survey (arXiv:2312.17044) mentions CoPE and Mamba twice each and none of the rest. The content-dependent half of the arena, the memory reading above and the navigation regime are in neither; the frequency-allocation line is in neither this note nor the positional review. The partitions are close to complementary. One survey does have the category: Liu et al.’s long-context survey (arXiv:2503.17407, March 2025) carries “content-aware position embedding”, with CoPE and DAPE as its members — the additive side, and before the multiplicative mechanisms arrived.

8 The relational map

Eight axes separate everything in this literature. They are not independent, but no two mechanisms below agree on all eight.

A1 Factor — magnitude (Re), phase (Im), both, or neither.

A2 Locus — instantaneous (c , a function of the current token) or accumulated (G , summed over the interval).

A3 Driver — the position index n , the token content x_t , or a query–key *pair*.

A4 Algebra — semisimple diagonal ($\mathbb{C}^\times$), unipotent (dual numbers, the shear), or non-abelian (GL).

A5 Sign — may the increment be negative?

A6 Rank of the content-to-increment map.

A7 Host — softmax attention, or linear attention / SSM.

A8 Composition — parallel prefix sum, path-ordered product, or neither.

Every *abelian* row above composes by **parallel prefix sum** (A8) — LieRE and Algebraic PE are the exceptions, since non-commuting generators satisfy no prefix-sum law and LieRE reads its rotation off a known position rather than accumulating anything. That is what being abelian buys, and it is the property the next table’s occupants give up.

mechanism	what breaks	consequence
PaTH, DeltaNet, RWKV-7 CoPE	the <i>fixed basis</i> (A4 non-abelian) <i>per-token factorisation</i> (A3 = pair)	path-ordered product of $I + \text{rank-1}$; no S_t $p_{ij} = \sum_{k=j}^i \sigma(q_i \cdot k_k)$; no θ_t
MesaNet, Titans, TAPE	the <i>path structure</i> (A8)	prefix-global inverse / deep memory / lay- erwise update
Shaw, T5, Transformer-XL	<i>continuity</i> (Puranik’s third)	a lookup table over clipped or bucketed lag; no generator M
TEM-t	<i>linearity</i> and <i>commutativity</i>	$e_t = \sigma(e_{t-1} W_{a_t})$; see §3.2

Table 1: Mechanisms that fit (24) or its unipotent extension. **Every row was read first-hand and every externally-checkable cell was checked against the source**, not against a summary of it; the sources are stored under `papers/` and indexed in `papers/INDEX.md`. *Jordan-RoPE’s magnitude factor is a *polynomial* in the lag, $d^r e^{i\omega d}$, not an exponential: its generator couples a complex eigenvalue to a nilpotent in one defective block, so it is non-semisimple rather than unipotent. [‡]FoX’s bias $\sum \log f_\ell$ and GLA’s weight $\prod \gamma$ are the same object; the unipotent label follows GRAPE’s realisation, which augments q, k to embed the bias in GL , and is a choice of embedding rather than a property of the mechanism. Read as $\mathbb{C}^\times$ it sits with Mamba/GLA. [§]Three cells are scoped rather than exact. LieRE’s $R(p) = \exp(\sum_i p_i A_i)$ reads its rotation directly off a known position rather than accumulating increments, and its A_i need not commute, so it obeys no prefix-sum law. Algebraic PE is $\mathbb{C}^\times$ for its sequence and grid constructions but non-abelian for k -ary trees, which need κ matrices. GRAPE’s additive family is “rank-1 (or low-rank)” in its own abstract and its phase-modulated instance emits one scalar, like FoX; “full” applies only to the general edge potential. [¶]MapEM evaluates the same equation on the *tensor product* of the content and position spaces; see §3.2.

mechanism	A1 factor	A2 locus	A3 driver	A4 algebra	A5 sign	A6 rank	A7 host
NoPE	—	—	—	—	—	—	softmax
RoPE	phase	acc.	index	$\mathbb{C}^\times$	n/a	n/a	softmax
ALiBi	magnitude	acc.	index	unipotent	fixed < 0	n/a	softmax
xPos	both	acc.	index	$\mathbb{C}^\times$	n/a	n/a	softmax
PoPE	magnitude	inst.	content	—	> 0	full	softmax
MapFormer	phase	acc.	content	$\mathbb{C}^\times$	signed	rank r	softmax
MapEM [¶]	phase	acc.	content	$\mathbb{C}^\times$	signed	rank r	softmax
MapPoPE	both	inst. + acc.	content	$\mathbb{C}^\times$	signed	rank r	softmax
Selective RoPE	phase	acc.	content	$\mathbb{C}^\times$	signed	full (or 8–16)	both
CARoPE	phase	acc.	content	$\mathbb{C}^\times$	$\in (0, 1)$	1/head	softmax
GRAPE-M	phase	acc.	index	$\mathbb{C}^\times$	n/a	n/a	softmax
GRAPE-A / AP	magnitude	acc.	content	unipotent	≥ 0	1-full [§]	softmax
FoX	magnitude	acc.	content	unipotent [‡]	< 0	scalar/head	softmax
Jordan-RoPE	both*	acc.	index	non-ss.	n/a	n/a	softmax
LieRE	phase	direct [§]	position	$SO(d)$, non-ab.	n/a	tile $b \times b$	softmax
Algebraic PE	phase	acc.	structure	$\mathbb{C}^\times$ [§]	n/a	n/a	softmax
Mamba / Mamba-2, GLA	magnitude	acc.	content	$\mathbb{C}^\times$	< 0	full/scalar	SSM
Mamba-3	both	acc.	content	$\mathbb{C}^\times$	signed	full	SSM

Reading the map. Three things fall out that no single row shows.

(i) *The frame’s own cells are full.* Crossing A1 (magnitude / phase) with A3 (index / content) gives four cells and all four are occupied: RoPE, ALiBi, Mamba-3’s rotation, FoX. GRAPE holds the index diagonal and the content-additive cell; Mamba-3 and Selective RoPE hold the content-phase cell. This is the finding of §6 in tabular form.

(ii) *A5 is varied deliberately — in a literature next door.* An earlier draft of this paragraph read “A5 and A6 are the axes nobody varies deliberately”. That is false, and it is worth spelling out how badly, because the correct version is more useful.

The *observation* is Sarrof et al. (arXiv:2405.17394), two years earlier and about SSM gates rather than positional encodings: “Across all non-time-invariant SSMs surveyed, we find that the gate is always **nonnegative . . . due to exponential or sigmoid parameterizations of the gate.**” That is the observation below — monotonicity as a squashing-function side-effect, never argued for — and it arrives with Theorem 2: such models cannot recognise PARITY. Grazi et al. (arXiv:2411.12537, ICLR 2025) then *prove* that finite-precision LRNNs whose state-transition matrices have only positive eigenvalues cannot solve parity, extend Mamba’s and DeltaNet’s range from $[0, 1]$ to $[-1, 1]$, and show the fix trains stably to 1.3B. PaTH adopts it on purpose (“we use $\beta_t \in (0, 2)$ as this allows for negative eigenvalues in the transition matrix, which has been shown to boost the state tracking performance”), and RWKV-7 restricts the range only because of observed training instability. **And Selective RoPE has already carried it into this exact cell:** its §4.2 cites Grazi, argues that “the input dependent rotations allow it to model *flips* depending on the input”, and reports that “Transformer with Selective RoPE is the only variant of Transformers capable of solving the parity task with a single layer up to this sequence length.”

What survives is narrower and is a statement about *reading*, not about novelty. GRAPE-AP needs $\omega = g(x) \geq 0$, CoPE’s gate is a sigmoid, and CARoPE’s $1/(\text{softplus} + 1)$ lands in the open interval $(0, 1)$ — so it can reverse neither *nor stay* — and none of those three cites Sarrof, Grazi, or negative eigenvalues at all (0 hits each). The axis is settled in the state-tracking literature and is not being applied as a design constraint in the positional-encoding-for-language literature. Two things follow that the corpus does not contain: nobody tests the axis on *navigation* rather than parity and formal languages (S_2 , A_3 , Dyck), and no comparison isolates it — the existing evidence is architecture-level, where $|\Delta|$ against Δ is one operation at identical parameter count. **Both are now done** (§10.2): $|\Delta|$ against Δ at identical parameter count, 12 seeds, on navigation. Monotone path integration beats an index code nowhere, and the learned code shows why — opposite actions fail to cancel at an opposition score of 1.85–1.98 out of a possible 2, against 0.11–0.13 signed. It is a replication in a new regime rather than a discovery, and SIGN_ABLATION_PREREG.md’s amendment says so in advance.

(iv) *A6 is thinner than a blank column suggests, but still open.*

Two rows constrain something rank-like for reasons of their own: PaTH’s Householder factors are $I + \text{rank-1}$ by definition, and CARoPE emits one scalar per head — so its content-to-angle map is rank 1 per head, *below* MapFormer’s r rather than above it. The nearest deliberate study is LieRE’s tile-size sweep, which varies density on the *output* side and peaks in the interior at 8×8 — the same shape of result as our step at $r = 2$, on the other side of the map. What remains unstudied is the *input-side bottleneck*: how many dimensions of content may steer the angle. That is the column where this project has a measurement — $r = 2$ is the worst setting, $r = 4$ buys +0.085 at $8 \times$ training length for 384 parameters, and at matched length the deficit is one of search rather than a skewed basis or missing capacity: a rank-2 solution exists and holds, and training finds it on 0/8 seeds (ours) and 2/8 (the paper’s per-head $r = 2$) against 8/8 at $r = 4$.

(iii) *A7 is nearly free.* Selective RoPE runs in both hosts and Mamba-3 crosses over by state-space duality, so “softmax versus SSM” separates implementations rather than mechanisms. It is

the axis that looks most important in the literature’s own framing and matters least in this one.

9 What each mechanism can and cannot do

The axes of §8 are descriptive. This section is the part that predicts: what a task demands of a positional code, which mechanisms supply it, and which trade-offs are forced rather than incidental.

9.1 Three hard trade-offs

(T1) Commutativity buys the scan and costs path-dependence. With $K = 1$ and a linear increment the phase depends on the multiset of tokens traversed, the prefix product collapses to a prefix sum, and inference is $O(\log T)$ parallel. The same property makes displacement-under-accumulated-heading unrepresentable: a fixed per-token increment cannot express “forward” when forward depends on where you are facing. Measured both ways — the position effect falls from $+0.438$ to $+0.050$ under turn/turn/forward actions, and a mechanism that abandons the group law (TEM-t, whose ReLU and non-commuting per-action matrices each break it independently) runs $120\times$ slower at $L = 2048$ than at $L = 128$ with $20\times$ fewer parameters. You may have the scan or the path-dependence.

(T2) A signed increment buys a map and gives up a clock. The accumulator’s growth exponent is set by how completely opposite actions cancel: $\alpha \approx 0.52$ (diffusive) when they do, $\alpha \approx 0.94$ (ballistic) when they cannot. But the exponent is a symptom; the substance is §9.2 — a signed increment makes the interval sum a *net displacement*, invariant to route and blind to elapsed time, and a monotone one makes it a *path length*, injective in time and blind to place. Neither dominates. On navigation the signed form wins outright (at matched training loss the monotone one does not beat a plain index clock anywhere); on language the monotone form is the correct object, and the signed one would be discarding the very distinction position exists to make.

(T3) Compressing the memory makes decay compulsory; keeping it makes decay optional and the cost quadratic. In the primal form the outer-product state saturates, so $\text{Re } G < 0$ is capacity management — something must evict. In the dual form items are kept separately and the softmax renormalises, so a decay term is an inductive bias one may decline. And a memory that must *overwrite* stale associations needs a delta-rule update, whose transition matrix is a reflection rather than a scalar — which forfeits T1.

9.2 Cancellation is not a quality axis: it chooses the question

The measurements above make it tempting to read cancellation as simply better. It is not, and the reason matters for choosing a mechanism.

Attention sees only the interval sum $\theta_s - \theta_t = \omega \sum_{t < u \leq s} \Delta_u$. What that quantity *is* depends entirely on whether the increments can cancel:

	Δ signed	Δ monotone
the interval sum is	net displacement	monotone in path length
two routes $A \rightarrow B$	agree	disagree
injective in	space , blind to time	time , blind to space
what collides	same place, different times	nothing — every instant is distinct
the accumulator is	a map	a clock

For a scalar accumulator these are mutually exclusive by construction (a rank- r accumulator can cancel in one subspace and count in another; audit 2026-09-10), and each is *correct* for one job. A cognitive map **needs** same-place-different-time to collide — that collision is exactly what revisit prediction asks for. Language needs the opposite: a code that made token 5 and token 15 interchangeable because some “backward” token intervened would be broken, not economical.

So the monotone increment of CARoPE, GRAPE-AP and CoPE is not an oversight that the state-tracking literature caught them in. **On language a monotone clock is the right object**, which is why RoPE — $\Delta \equiv 1$, maximally monotone — is the sensible default there and why nothing went visibly wrong. The limitation appears only where the task demands a quantity that is invariant to route, which on text is essentially parity and formal-language state tracking, and on navigation is everything.

The growth exponent is a readout of which question, not a score. $\alpha \approx 1$ means the accumulator is measuring elapsed time, which is unbounded, so the code must eventually leave any range it was trained on. $\alpha \approx 0.5$ means it is measuring a random-walk position. The ideal for a bounded map would be $\alpha = 0$ — position on a torus is bounded — and nothing here achieves it, because every mechanism accumulates without wrapping. It is not a design gap either (corrected, see below): wrapping is a no-op, since θ enters only through $\cos, \sin$, and any other bound breaks additivity.

What MapFormer gave up to get the map. A signed Δ makes two visits to the same cell positionally identical. That is the point, and it also means the phase cannot represent *how long ago*. Anything recency-dependent has to get that information from somewhere else.

Which is a candidate account of the forget gate, listed elsewhere as mechanism-unidentified. It adds a second, content-dependent, *monotone* accumulator in the magnitude slot — measured at $\alpha = +0.956$, ballistic, beside the phase accumulator’s 0.6 — so it restores the clock that the signed phase gave up. It would account for every otherwise-puzzling detail (untested; the batch registered to test it was deleted before analysis): it needs a *live* λ (frozen at zero it lands on the baseline, -0.016 , unmeasured) but not *decay*; 5 of 8 seeds learn $\lambda < 0$, and those gain most ($r(\lambda, \text{gain}) = -0.516$). A negative λ still gives a monotone accumulator, counting up instead of down. On this reading it was never a forget gate — it is a clock, and the sign of λ only sets which way it runs.

Testable, and untested: any monotone content-dependent increment of either sign should reproduce the gain; a content-*independent* constant increment (a pure index clock) should reproduce much of it; and the gain should vanish on a task with no recency structure.

The growth law is an association, not a cause — and my proposed test of it was malformed (2026-09-07). I wrote that the direct test is to “vary α deliberately and check that OOD degradation follows”. It cannot be done as stated, and noticing why is worth more than the experiment would have been.

α is not a parameter of any model here; it is a statistic fitted to a trained one, and nothing in the code is raised to it. The two ways to move it both fail. Interpolating signed $\rightarrow$ monotone ($\Delta_\lambda = \lambda\Delta + (1 - \lambda)|\Delta|$) does move α from 0.52 to 0.94, but it destroys the map in the same stroke, so on a map task it isolates nothing the sign ablation did not already measure. Bounding the accumulator while keeping the sign is either a *no-op* — θ enters through $\cos, \sin$, so θ and $\theta + 2\pi$ are already identical — or, if done by squashing, it stops $\theta_s - \theta_t$ being the interval sum and destroys interval-relativity.

So within this frame α may not be independently controllable: it is set by the cancellation, whose only levers are A5 and A6. That would make it a re-description of those two axes rather than a third cause, which is exactly consistent with its covering those two mechanisms and none of the other four.

And there is a question prior to the experiment that I had skipped: **why should a growing accumulator cost anything at all, when the code is periodic?** The standard answer is the critical-dimension account — low-frequency channels read at untrained phases — which was imported and **refuted** here: ablating them costs MORE at extrapolated length, not less. So the association is established ($r(\text{opposition}, \alpha) = +0.9995$, and α tracks both the sign and rank effects) and the route is open. This is the same class of error as H1 in the recency pre-registration: a well-sounding test that could not have discriminated.

Tested directly 2026-09-07, and it holds — with two of my own hypotheses refuted. Until this test every task measured here wanted a *map*, so “signed wins” and “signed suits this task” were not separable. On a k-back retrieval task with uncounted filler (‘RECENCY_RESULTS.md’), six arms $\times$ eight seeds in one batch:

- **The crossover.** Constraining the increment to be monotone costs -0.280 raw / -0.215 loss-matched on the torus (12/12 seeds) and -0.004 raw / $+0.026$ loss-matched here, both inside their MDE. Interaction $\approx +0.28$.
- **α is diagnostic.** The unconstrained arm learns $\alpha = 0.591$ on the torus and 0.967 here ($\Delta = +0.376$, se 0.009), while every constrained arm sits at ≈ 1.0 on both — the control that attributes the shift to the choice.
- **Mechanism, by intervention.** A learned content gate, 8/8 seeds per head. Correlation could not settle it ($r(\text{gate}, \text{acc}) = -0.34$). Magnitude-matched pair: a constant Δ on content scores 0.783 , on every token 0.189 ; $+0.594$ at 8/8.
- **Index collapses,** 0.234 against 0.96 – 1.00 ($+0.750$, 8/8), larger than the navigation effect. This is CoPE’s published claim reproduced, not a new one.

Scope (audit 2026-09-10 #2, recorded 2026-09-25). The recency task does not *need* a clock: a signed rewind of the query token solves it exactly (a hand-set kernel selects the answer 1423/1423). So the crossover shows that cancellation can be given up there at no cost, not that the task demands a clock, and nothing inverts. The two costs are also read at different extrapolation ratios: the torus at $8\times$ the training length, recency at $2\times$.

Refuted, both mine, both recorded before the batch or immediately after. (i) I predicted an index code would do WELL on a clock task; a *contextual* clock is precisely what a fixed index cannot encode, and the sign inverted. (ii) I predicted monotone would BEAT signed here. That is malformed: the signed arm is unconstrained and *contains* the monotone solution, so signed $\geq$ monotone on any task up to optimisation — the same defect as the sign ablation’s unmeasurable discriminator, in a different disguise. The well-posed form is the cost of constraining, above.

A confound I nearly published. The condition I pre-stated as decisive — make the filler count — collapses to 0.086 . But a control changing *only* θ ’s scale, still counting content alone, collapses just as hard (0.110). The model is acutely sensitive to θ ’s magnitude, so that condition proves nothing. The magnitude-matched pair was added afterwards.

Read this way the frame’s two slots are not two knobs but **two answers**: a signed phase is a map, a monotone accumulated magnitude is a clock, and a system with a scalar accumulator that needs both must spend both slots (a rank- r phase can in principle cancel in one subspace and count in

another). Selective RoPE and Mamba-3, which both pair a data-dependent rotation with decay, are then the natural endpoint rather than a curiosity, and the empty diagonal of §8 — content-dependent phase *and* content-dependent magnitude, in softmax attention — is the cell worth building.

9.3 What a task demands, and what supplies it

if the task needs ...	it requires	and these supply it
<i>order only</i> — which token came first	any monotone clock	RoPE, ALiBi, CoPE, CARoPE, GRAPE-AP, every gated linear RNN
<i>signed displacement</i> — parity, net motion, state tracking	a signed increment	MapFormer, Selective RoPE, Mamba-3; DeltaNet/PaTH/RWKV-7 with $\beta > 1$. Not CARoPE, GRAPE-AP or CoPE
<i>absolute location in a bounded map</i>	signed, and rank $r \geq 4$ per head to be found reliably	MapFormer at $r \geq 4$. At $r = 2$ (ours shared or the paper’s per head) training usually fails to find the torus solution within budget, though a rank-2 solution exists: search, not a skewed basis or capacity
<i>holding position far past the training length</i>	a diffusive accumulator, $\alpha \approx 0.5$	follows from cancellation (T2), as an association; a wrapped filter does <i>not</i> bound the accumulator (it wraps the innovation: range 285.6 against 283.9)
<i>path-dependent actions</i> (turn/forward)	non-abelian transport, or an allocentric recoding of the input	PaTH, DeltaNet, RWKV-7, TEM-t — none evaluated on navigation. Recoding the action to absolute displacement restores +0.488 and keeps the scan
<i>overwriting stale associations</i>	a delta rule	DeltaNet, PaTH, RWKV-7; forfeits the parallel scan
<i>exact recall over a window</i>	the dual form; keep the items	softmax attention. A lossy summary is not a sufficient statistic, which is why hierarchy loses here; on compositional transfer it is ahead in direction only (unpowered at $n = 8$)
<i>bounded state, linear time</i>	abelian transport	every row of Table 1; measured $2.6\text{--}3.3\times$ over a $16\times$ length increase

9.4 What follows for choosing a model

On language, most of this is invisible. Order is nearly all that is demanded, so a monotone clock suffices, which is why the constraint has gone unnoticed on that side: CARoPE, GRAPE-AP and CoPE are all monotone and none cites the state-tracking literature that proves the limitation. The one place it surfaces on text is parity and formal-language state tracking, where it has been both proved and fixed.

On navigation and any task with a metric structure, the sign and the rank of the increment matter most per parameter. They are worth more per parameter than anything else measured here — the sign costs nothing at all, and $r = 2 \rightarrow 4$ costs 384 parameters. They do not dominate the choice of positional encoding: the “two orders of magnitude” once written here (+0.461 against +0.003) was a 16-epoch recipe, and converged the encoding main effect is +0.189 at $T = 1024$, larger than rank’s +0.085.

Where the environment breaks the assumption, fix the environment. No abelian mechanism can

represent turn/forward dynamics, and the non-abelian ones pay the scan for it. Recording the absolute displacement instead of the commanded turn is a change to the data, not the model, and it restores the full effect — available wherever heading is known, which is every simulator.

The two literatures are complementary, and the empty cells are informative. The fast-weight line owns the magnitude slot and never touches the phase; the positional line owns the phase and treats decay as a side-condition. Selective RoPE and then Mamba-3 claim both. A mechanism that needs a bounded state *and* signed transport *and* rewritability does not currently exist, and T1 says the last two cannot both be had with a parallel scan.

10 What is measured

10.1 What the constraints are worth

The first four rows are one batch of 6 arms (`_SELECTIVE_TORUS.json`); the r row is a second batch (`RANK_SWEEP.json`). They are comparable because the shared **Vanilla** arm is bit-identical across the two at every length ($\max |\Delta| = 0$). Increments are against MapFormer at $r = 2$; MDE is $2.8 \text{sd} / \sqrt{n}$.

constraint removed	params	parity $L=128$	torus $T=1024$ (trained 128)
$K = 1 \rightarrow 4$ (causal conv)	+193	−0.020 detectable	−0.064 ($t -1.99$)
$m = r \rightarrow Hn_b$ (full rank)	+7,873	−0.020 detectable	+0.058 (7/8)
φ linear $\rightarrow$ gated	+8,193	−0.030 detectable	+0.086 (8/8, sign $p=0.008$)
all three (Selective RoPE)	+16,385	−0.009 unmeasured	+0.048 ($t 1.36$)
$r = 2 \rightarrow 4$, <i>constraint kept</i>	+384	—	+0.085 (8/8, $t 3.57$)

CONFOUND — found on one audit, then *downgraded* by the next. The first audit recorded that these arms are not single-knob: **SelectiveAngle** carries no ω buffer, so every row also replaces the readout $\text{diag}(\omega)W_{\text{out}}$ by τW_2 , discarding the 64 geometrically-initialised frequencies spanning $[2\pi/64, 2\pi]$ that §2.5 calls the thing making the phase a multi-scale code rather than 64 unrelated clocks.

That reading is too strong, and the code says so. The two rows it matters for — $K=1 \rightarrow 4$ and the gate — are built with **RANK=2**, i.e. $\text{proj} = \text{Linear}(d, 2) \rightarrow \text{Linear}(2, C)$, both bias-free, followed by $\tau \cdot \text{cumsum}$. Since $\text{diag}(\omega)W_{\text{out}}$ and τW_2 range over the same set of $C \times r$ matrices, **the function class is preserved exactly** — it is the construction of §2.4, and `verify_inclusion.py` already certifies it. What actually differs is the *initialisation* (geometric ω against a default **Linear** draw) plus one scalar τ .

So the caution stands but changes kind: these are **initialisation**-confounded, not structurally confounded, and the conv / gate effects are attributable up to init. The two full-rank rows ($m = r \rightarrow Hn_b$ and all-three) are a separate matter — they also add **weight_norm** and a bias, so their class genuinely differs. The arm that would settle it — geometric ω retained, one knob added — has still not been run. The parameter deltas are unaffected and verify exactly.

Three readings, in decreasing order of how well they are supported.

Rank is the only axis with a measurable effect, and it saturates at 4. $r = 4, 8, 16, 32$ are flat within noise (+0.085 to +0.095), so this is a step at $r = 2$, not a capacity slope. The same gain is available inside MapFormer’s parameterisation for $21 \times$ fewer parameters than outside it.

The gate is not doing what its name suggests. It is statistically indistinguishable from the unrelated full-rank addition of comparable size (+0.018 at $T=512$ against an MDE of 0.026), so the parameters and not the mechanism are the plausible cause. The natural mechanism — that a

sigmoid suppresses observation tokens, which MapFormer must instead learn to send to $\Delta \approx 0$ — was tested directly and falsified: the gate is $1.35\times$ larger on actions than observations on the torus *where it helps*, and $1.54\times$ larger on the analogous split on parity *where it hurts*.

The convolution is the one knob with a principled sign, and it is negative on both tasks at negligible cost. With $K > 1$ the increment of a token depends on its neighbours, so the phase ceases to be a function of the multiset of tokens traversed and the map from action sequences to $SO(2)^{n_b}$ is no longer a group homomorphism. Commutativity is MapFormer’s entire computational argument — and the word does not appear once in Selective RoPE. The measured cost of giving it up is -0.020 on parity and -0.064 on the torus for 193 parameters.

10.2 The sign of the increment (A5)

The one axis of §8 where every mechanism outside MapFormer sits on the same side. Six arms $\times$ 12 seeds, one batch, clean torus, $r = 4$ throughout, **parameter count identical at 204,757** — the constraint adds nothing and does not touch the rank. Pre-registered in SIGN_ABLATION_PREREG.md, including the verdict for the outcome that occurred.

arm	Δ	$T=128$	$T=512$	$T=1024$	final loss
Signed	$W_{\text{out}}W_{\text{in}}x$	1.000	0.978	0.922	0.0002
Vanilla	(RNG control)	1.000	0.984	0.927	0.0002
Abs	$ W_{\text{out}}W_{\text{in}}x $	0.946	0.675	0.558	0.171
Pos	softplus($\cdot$), GRAPE-AP	0.977	0.798	0.584	0.068
CARoPE	$1/(\text{softplus}(\cdot) + 1)$	0.900	0.809	0.645	0.293
RoPE	index	0.799	0.449	0.345	0.784

The result, and it does not depend on the accuracy table’s caveats. At *matched training loss*, signed path integration beats an index code by $+0.123$ and $+0.195$ at $T = 512$ and $T = 1024$, 12/12 seeds. Monotone path integration beats an index code **nowhere**: -0.028 at $T = 128$ (detectable) and unmeasured beyond. *The entire measured value of a content-dependent phase over a plain index clock, on this task, requires the increment to be signed.*

Mechanism, at the level of the learned code. The opposition score $\|\Delta(+x) + \Delta(-x)\|/\|\Delta\|$ is 0 when opposite actions cancel exactly and 2 when they are identical:

arm	oppose _x	oppose _y	$ \cos(N, E) $
Signed / Vanilla	0.125 / 0.128	0.106 / 0.130	0.218 / 0.133
Abs	1.849	1.855	0.587
Pos	1.905	1.885	0.723
CARoPE	1.981	1.977	0.930

CARoPE’s published parameterisation reaches 1.98 of a possible 2.00, with its north and east axes all but collapsed onto one another. These models do not merely score worse — they **cannot represent a -1 action**, and the probe shows they do not.

The pre-registered accuracy discriminator failed, by my design error. It asked for a deficit at *training length* as well as OOD, on the reasoning that a representational failure should appear where the code is formed. At $T = 128$ the baseline is 1.000 ± 0.000 : 0.054 of headroom against an MDE of 0.057. The cell cannot show a deficit of any size, so the test required something the design could not deliver — the rule-11 ceiling trap, written into my own verdict rule.

The training-length effect is nonetheless there, in the *loss*: every constrained arm fits strictly worse than the signed baseline on 12 **of** 12 **seeds** at identical parameters (+0.171, +0.067, +0.293 against 0.0002). Loss-matching at $T = 128$ therefore partials out the very quantity the constraint causes, and $r(\text{loss}, \text{acc}) = -0.978$ there in any case. A ceiling on accuracy is not the absence of an effect; look at what the model could not fit.

Scope: a replication in a new regime, not a discovery. Recorded before the first checkpoint was read. Sarrof et al. observed the non-negativity and proved it blocks PARITY; Grazzi et al. proved the eigenvalue version and fixed it; Selective RoPE’s §4.2 already demonstrated single-layer Transformer parity from rotations that “model *flips*”. New here is the regime (navigation, not parity or formal languages) and the isolation — $|\Delta|$ against Δ is one operation at identical parameter count, with an RNG-path control that lands at +0.006, unmeasured at every length.

10.3 A prediction, and the paper’s own 5D negative

MapFormer v4 Table 6 reports MapWM collapsing in 5D: 0.75/0.50/0.35 on IID/OOD-d/OOD-s, *below* CoPE at 0.94/0.80/0.69, while MapEM-s holds at 1.00/1.00/0.87. *Corrected on audit, twice.* First, the paper does give an account, and it is a *neuron-budget* account, not a packing one: “performances quickly degrade, since the amount of **dimensions** to represent **with a fixed neuron budget** grows exponentially” (Table 6 caption), glossed in the body as “the difficulty of encoding each axis accurately in high dimension, given a fixed neural budget”. An earlier draft printed “positions” for “dimensions” and dropped the budget clause, which converted their explanation into the one below. Its 5D setting is **grid size 5 at sequence length 16**, against the reproduction below at grid 4 and $T = 128$.

And second, the premise of what follows is counterfactual. §D.1 states the rank outright: “for each model, we set the dimension of its inner rank to the dimension of the world, i.e.: $\Delta_{\text{in}} \in \mathbb{R}^k$, $k = 3, 5$.” So their 5D run uses $r = 5$, not $r = 2$, and the 5D collapse cannot be a rank artefact of the kind hypothesised here. That also agrees with §2.5’s own reading of App. A.7 (r is the dimensionality of the action space) and contradicts the “for instance $r = 2$ ” gloss, which in the source is specific to 2D: “in 2D, Δ_{in}^t is the 2D movement vector ... where $r = 2$ ”. The subsection is kept because the experiment it motivated was run and reported below, and because its result stands on its own; the motivation does not.

Equation (15) supplied the candidate. Had $r = 2$ been held fixed at $D = 5$, the phase code would be confined to a 2-torus while the task needs a 5-torus, the separation exponent would be $D/r = 2.5$ rather than the floor value 1, and position becomes progressively unresolvable as the map grows. MapEM’s survival is consistent on the same reading: its multiplicative $A_X \odot A_P$ lets the content channel disambiguate what a degenerate position channel cannot.

How strong this is, stated plainly. It is weaker than the kernel claim it replaces, in the way that a correct argument is often weaker than a wrong one. (15) predicts *degradation growing in D/r and in N* , not an impossibility, so it cannot say on its own whether 5D falls to 0.35 or merely slips. And their r in 5D is not observable from the paper. The strongest available claim after the experiment below is “ $r = 2$ suffices to produce a 5D collapse of the reported shape in our reproduction”, which is a candidate explanation for their number and not a demonstration of it.

Design. A D -dimensional torus (`environment_nd.py`): 2D actions $\pm e_i$, allocentric aliased observations, the interleaved stream and revisit-masked loss of the paper’s task, and — mirrored exactly, because it sets the revisit statistics every 2D number here rests on — the *directed* walk, which

draws an action and repeats it $k \sim U\{1..10\}$ times. State-space size is held near-constant so that only dimensionality moves: $D = 2$ at grid 32 (1024 cells), $D = 3$ at grid 10 (1000), $D = 5$ at grid 4 (1024). Arms $r \in \{2, D, D+2\}$, 8 seeds, one batch, the recipe of the 2D rank sweep.

What the two mechanisms predict, separately. The scale-free minimum separation at each cell of the design, from the gate:

	$D = 2, N=32$	$D = 3, N=10$	$D = 5, N=4$
$r = 2$	0.047	0.033	0.020
$r = D$	0.047	0.065	0.218
$r = D+2$	0.067	0.193	0.388
ratio $r=D : r=2$	1.0×	2.0×	11×

- **Geometry** predicts the $r=2$ deficit ordered $D=5 \gg D=3 > D=2$, tracking the ratio row, and predicts *nothing* for $r = D$ versus $r = D+2$, which are both above the threshold.
- **Optimisation** predicts a further gap from $r = D$ to $r = D+2$ of roughly the size measured in 2D (≈ 0.085 at $8\times$ training length), present at every D and *not* growing with it.

The two are separable because they act on different cells of the same table, which is the reason for running all three ranks rather than two.

RESULT ($n = 8$, **one batch**, DXR_RANK_THRESHOLD.md).

$T = 512$	$D = 2$ (grid 32)	$D = 3$ (grid 10)	$D = 5$ (grid 4)
$r = 2$	0.848 ± 0.056	0.707 ± 0.077	0.896 ± 0.076
$r = D$	—	0.793 ± 0.092	0.877 ± 0.072
$r = D+2$	0.959 ± 0.036	0.860 ± 0.055	0.950 ± 0.045
$r=D+2$ minus $r=D$	+0.110	+0.067	+0.073

The geometric half is refuted and the optimisation half stands. Two of the three falsifiers below fired. $r = 2$ is not impaired at $D = 5$; it posts its best score of the three conditions there. And its deficit does not grow with D : at $T=512$ it is +0.110, +0.153, +0.055 for $D = 2, 3, 5$ — *smallest* where the bound says it should be largest. Meanwhile the $r=D \rightarrow r=D+2$ gap is +0.110/ +0.067/ +0.073 at $D = 2/3/5$ and does not grow with D . *Corrected on audit*: paired at $T=512$ these are DETECTABLE at $D=2$ (t 4.08, 7/8) and $D=5$ (t 3.35, 8/8) but **unmeasured** at $D=3$ (t 1.93, 6/8); and at $D=2$, $r=D$ is $r=2$, so that cell is numerically the same contrast as the $r=2$ deficit and cannot discriminate the two hypotheses. The optimisation half rests on **one** independent detectable cell, not three.

The account of Table 6 offered above is therefore WITHDRAWN. The paper’s 5D collapse is not explained by $r = 2$; in this reproduction $r = 2$ handles $D = 5$ better than it handles $D = 2$. Whatever causes their collapse, rank deficiency of the position code is not it.

Rule 9 flag on one cell. At $D = 2$, $T = 128$, $r(\text{final loss, accuracy}) = -1.000$ over 16 runs, so that column carries no information the training loss does not and its +0.015 must not be read as an architecture effect. The $T=512$ columns are informative ($r = -0.70/ -0.75/ -0.18$).

Falsifiers (as pre-registered). $r = 2$ unimpaired at $D = 5$ kills the geometric account entirely and with it the explanation of Table 6. A flat $r=D$ versus $r=D+2$ contrast at every D makes the optimisation half a 2D artifact. A deficit at $r = 2$ that does *not* grow with D leaves both halves standing but refutes (15) as the mechanism, since the exponent is the only thing D changes.

Gates, run before any GPU time (ND_GATES.md). All three configurations pass: an n -gram on the action stream alone at orders 1–5 reaches at most 0.536 against a measured majority-class rate of 0.526, and revisit rates are 0.23/0.28/0.62 with 30–79 scored positions per trajectory. The revisit gate is not a formality here: a simple random walk on $\mathbb{Z}^D$ is *transient* for $D \geq 3$ (Pólya), so revisits exist only because the torus is finite, and a configuration where they collapsed would show a “rank effect” that was really an absence of labels. The cross- D difference in revisit rate is a real confound and is the reason the primary contrast is *within* D ; cross- D comparisons here are descriptive only.

10.4 The difference that has not been tested

Everything above concerns the *generator*. Selective RoPE also differs in *placement*: it derives the angle per head and per layer from that layer’s own (ℓ_2 -normalised) input — their ablation moved it there from the queries, see the scope box under Selective RoPE in §2 — whereas MapFormer computes θ once from the token embeddings and reuses it in every layer. (θ is per head in both; in this implementation only MapFormer’s r -dimensional latent is shared across heads.) The arms here swap the generator only, and at one layer the distinction is nearly vacuous, since $q = W_Q \text{LN}(x)$ is itself a learned linear map of the token. At depth it is real, and it is the sole axis on which the two designs remain uncomparated.

It is also not the axis probed by the refinement experiment of Appendix A: that carried a *bounded gated correction* to θ across iterations of a shared block and the optimiser declined to use it. Recomputing θ from scratch at each layer, from that layer’s representation, is a different model and an open question.

10.5 Empirical anchors

mechanism	sets	parameter cost	measured effect
Path integration	phase θ	+448 on 199,042 (+0.23%)	+0.078 parity (16/16); +0.461 torus (16-epoch recipe; converged +0.243 at $T=128$)
PoPE	magnitude μ	$d_{\text{head}}/2 \rightarrow d_{\text{head}}$ widening, +320	+0.027 <i>with</i> path int. Without it, both arms were on the floor at 16 epochs — see note
Bottleneck rank	state rank ρ	+384 on 204,373 (+0.19%)	+0.085 torus $T=1024$ for $r=2 \rightarrow 4$ trained at 128 (8/8); flat to $r=32$; trained at 1024, solved 8/8 against 0/8
Recursion	block reuse	0	+0.056 parity (16/16); matches or beats $3\times$ params
Hierarchy	token count	0 if weight-shared	length-dependent: ≈ 0 at $L = 16$, +0.060 at $L = 512$ (23/24 seeds); -22.8% time at $L = 2048$, but see note (iv)
InEKF	$\theta \rightarrow \hat{\theta}$	+49,600 on 204,373 (+24%)	no individual <i>component</i> is load-bearing; the whole filter is +0.06 to +0.13, at OOD length only
Signed Δ	sign of the increment	0	the largest effect here per parameter. Removing it: $-0.215 / -0.280$ loss-matched at $T=512/1024$ (12/12); a monotone phase beats an index clock <i>nowhere</i>

Reading the table. Parity figures are at $L = 128$ trained at $L = 16$, from one batch of $8 \text{ arms} \times 16$ seeds. Torus figures are the position-axis factor average over both encodings (+0.4379 RoPE, +0.4851 PoPE). **Three qualifications the columns cannot carry.** (i) The hierarchy row is *not* a parity-at- $L = 128$ figure: its accuracy comes from runs trained at $L = 16$ and $L = 512$, and its time saving is measured at $L = 2048$ — at $L = 128$ the same benchmark shows hierarchy alone at +2.0% (and the combined loop $\times$ hierarchy arm at +0.4%), i.e. no saving at all. (ii) The InEKF’s parameter cost is measured on a 204,373-parameter torus model and the path-integration cost on a 199,042-parameter parity model; the two percentages are not commensurable, and the InEKF has never been run on parity. (iii) PoPE without path integration is a difference between two models sitting on the measured floor of 0.506 (0.509 vs 0.530), with a seed spread wider than the gap — it is a floor artifact, not an effect, and is omitted from the column for that reason. *Corrected 2026-09-25:* the floor was the 16-epoch recipe; converged, PoPE on the index row is -0.126 at $T = 128$ and ahead past it (PAPER2X2_RESULTS.md). (iv) *Corrected on audit:* an earlier version of this note claimed the -22.8% bundled the shared block’s saving with the pooling’s. It does not. LOOP_HIER_COMPUTE.md isolates both axes at exact parameter parity at $L=2048$: hierarchy alone (HOURGLASS_K2 vs HOURGLASSFLAT3, both 596,034) is -22.9% time and -19.9% memory, while sharing alone (LOOPEDHOURGLASSFLAT vs HOURGLASSFLAT3) is +0.1%. Attributing -22.8% to the hierarchy axis is accurate.

A The other mechanisms, in the same notation

Weight-shared recursion (the “loop”). With B_θ a single transformer block and n the loop count,

$$h^{(0)} = \text{Emb}(x), \quad h^{(j+1)} = B_\theta\left(h^{(j)}; \cos \theta, \sin \theta\right), \quad j = 0, \dots, n-1. \quad (28)$$

θ is computed *once* from the token embeddings and reused, matching both parents. Parameter count equals one layer; compute equals n layers.

Carrying a *bounded, gated correction* to θ across iterations — $\theta \leftarrow \theta_0 + g \cdot \tanh(\text{refine}(h))$ with g initialised to zero — is a different model, and it is null. On the action-noise run the null is drawn from ($n = 9$), the learned gate settles at mean $|g| = 0.144$, max 0.320, with the sign *consistent* in 7 of 9 seeds — so the correction is capped at 0.32 rad against a θ spanning $\sim 2\pi T$, i.e. the optimiser points the refinement somewhere and then makes it negligible.³ The gate was verified escapable (gradient 1.9×10^{-3} at zero), so the optimiser declined rather than failed to express it. A genuine per-iteration recomputation of $\omega \cdot \text{cumsum}(\Delta(h^{(j)}))$ has *not* been run.

Level 1.5 invariant EKF (withdrawn; recorded for completeness). A correction applied to θ before the rotation:

$$R_t = \exp(\text{clamp}(f_R(x_t), -5, 5)) \quad \text{per-token measurement noise} \quad (29)$$

$$z_t = \pi \tanh(f_z(x_t)) \quad \text{measurement in } [-\pi, \pi] \quad (30)$$

$$\nu_t = \text{atan2}(\sin(z_t - \theta_t), \cos(z_t - \theta_t)) \quad \text{wrapped innovation} \quad (31)$$

$$K_t = \Pi / (\Pi + R_t), \quad \Pi = e^{\log \Pi} \quad \text{gain; } \Pi \in \mathbb{R}^{H \times n_b}, \text{ constant in } t \quad (32)$$

$$d_t = (1 - K_t) d_{t-1} + K_t \nu_t, \quad d_{-1} = 0 \quad \text{affine scan} \quad (33)$$

$$\hat{\theta}_t = \theta_t + d_t. \quad (34)$$

Π is a learned ($H \times n_b$) constant — 64 values at the default config, one per head and frequency block — not a scalar; it is constant across *tokens*, which is what distinguishes Level 1.5 from the full heteroscedastic filter. The recurrence is scalar in each channel and affine, so it is one more parallel scan.

Initialisation, which turned out to matter. $\log R$ ’s head is initialised with weights scaled by 0.01 and bias 0, giving $R \approx 1$ and hence $K \approx 0.5$ *at initialisation* — corrections at half strength before anything is learned. On the multiplicative-attention backbone this destroyed the gradient path and diverged 3 of 9 seeds; the fix was a bias of 3.0, i.e. $K \approx 0.05$, making the filter a near-no-op at init.

Status. No individual component is load-bearing: the measurement head, the per-token gate and the learned Π can each be removed alone at no measurable cost ($n = 5$). The whole filter does have an effect, but only at out-of-distribution length (+0.062 at $T=512$, +0.124 at $T=1024$, *loss-matched only* — the raw contrasts are +0.057 and +0.120, both inside their MDEs of 0.150 and 0.180 at $n = 5$, so the detectability comes entirely from the post-hoc regression on final training loss), and on a task built to supply the drift it is meant to correct the effect *does not grow with the drift* — flat in one recipe, negative in another. The mechanism is withdrawn as inference; what survives is stabilisation.

³An earlier draft gave $|g| \approx 0.083$ with *inconsistent* sign. Those are the Match-Query checkpoints — the premise-invalid run — read by mistake for the action-noise ones. The conclusion is unchanged and the supporting reading is not: the sign is largely consistent, so the argument rests on the magnitude alone.

Hierarchy (hourglass). With pooling factor k , a causal *shift-then-mean* pool P_k and an upsample U_k ,

$$h \leftarrow B_{\text{pre}}(h), \quad h \leftarrow h + U_k B_{\text{coarse}}(P_k h), \quad h \leftarrow B_{\text{post}}(h), \quad (35)$$

with the coarse angle pooled alongside the content, and the sequence padded to a multiple of k then truncated after B_{post} .

Two details the notation would otherwise hide. P_k right-shifts by $k - 1$ with a zero pad *before* averaging, so coarse token j is the mean over original indices $jk - k + 1, \dots, jk$ — not the plain group mean, and the first coarse token is dragged toward zero by the pad. And U_k is `repeat_interleave`, which is *not* the adjoint of P_k (that would carry a $1/k$ and the matching shift); it is written U_k rather than $P_k^\dagger$ for that reason.

Setting all three blocks to the *same* shared module gives the weight-shared hourglass: one block’s parameters (204,373, identical to the flat 1-layer model, against 600,917 unshared) with the coarse block’s token reduction retained. Note that every hourglass class ignores `n_layers` and is always this 3-block scaffold, so a flat control must be built at 3 blocks, not 1.